Climate Conditions and Flooding Disasters in Nova Scotia

Data Analysis STAT 4620/5620 Winter 24-25

Fatma Sarhan, Margo Paris, Thu Nguyen, Kyle Allie

Project's Github Repository

github.com/KyleA2001/STAT4620_5620-Flooding_Project.git

Abstract

Summary in 150 words (Best done last, summarizes the whole project)

Keywords

- Storm Surge
- Autocorrelation Function (ACF)
- Cross-Correlation Function (CCF)
- Generalized Additive Mixed Models (GAMMs)
- Vector Autoregression (VAR)
- Granger-Causality
- Extreme Value Analysis (EVA)

1. Introduction

Coastal communities are increasingly vulnerable to the impacts of climate change, as rising sea levels and more frequent extreme weather events intensify the risks of flooding, infrastructure damage, and ecosystem disruption[1]. In urbanized coastal areas, population density and development have reduced natural protections, increasing exposure to geophysical hazards[2]. Nova Scotia, with its extensive Atlantic coastline, is no exception-experiencing a notable increase in heavy rainfall, strong winds, and severe flooding over the past decade[3]. Therefore, recognizing the drivers of coastal flooding in the region is crucial, especially as climate change affects atmospheric and oceanic conditions[4].

A key contributor to coastal flooding is storm surge which is an abnormal rise in sea level caused primarily by high winds and low atmospheric pressure during severe weather events[5]. While wind speed is widely recognized as the dominant factor influencing surge height, other variables such as atmospheric pressure, sea surface temperature, and precipitation also play important roles[6]. For instance, warming ocean temperatures have been linked to stronger storm systems and more intense surges[7], whereas rainfall, though critical to overall flood risk, tends to have a weaker direct relationship with surge dynamics[8]. Understanding the combined

[1] Helderop and Grubesic (2019)
[2] Von Storch and Woth (2008)
[3] Childs (2017)
[4] Simonović (2012)
[5] McInnes et al. (2003)
[6] Resio and Westerink (2008)
[7] Needham, Keim, and Sathiaraj (2015)
[8] Fang et al. (2021)

influence of these factors is essential for improving flood prediction models and enhancing coastal resilience.

Modeling these extreme weather events is challenging due to their rarity and the complex interaction of multiple climatic factors. Although flooding events may occur infrequently, their impacts are severe-causing significant societal and economic damage. This highlights the importance of developing more accurate analytical and forecasting tools. The goal of this research is to investigate the relationship between various climatic variables—specifically wind speed, temperature, and rainfall—and storm surge levels at selected locations in Nova Scotia, specifically Halifax and Yarmouth. The study is guided by the following research questions:

1. What is the relationship between various climatic factors and water surge levels across Nova Scotia?
2. How can the most significant correlated factors predict future flooding catastrophes in Nova Scotia?

By identifying key predictors of storm surge, this research aims to enhance our understanding of how climatic factors contribute to flooding disasters and supports the development of improved forecasting models, helping communities mitigate the risks associated with extreme coastal flooding.

2. Data

2.1 Description

The data used in this study were compiled from multiple sources, including Environment and Climate Change Canada[9], and the Government of Canada[10]. Specifically, water level data were retrieved from the Government of Canada, while wind speed, temperature, and rainfall data were sourced from Environment and Climate Change Canada.

Using the collected water level measurements, storm surge levels were derived through a pre-processing step, which is detailed in a later section.

Data were collected for two coastal locations:

- Halifax: From January 1961 to August 2014.
- Yarmouth: From May 1956 to December 2023.

Due to limitations in the available data within the examined time frame and insights from other researchers, three covariates were selected for analysis: air temperature, wind speed, and rainfall. A description of these covariates is provided below:

[9] weatherstats.ca (2024)
[10] Fisheries and Oceans Canada (2024)

Variables	Units	Description	Frequency
Date	days	Date when data was recorded	Daily
Location		Halifax and Yarmouth	
Water Level	m	Height above datum or reference system	Hourly
Storm Surge	m	Water level - R calculate tidal component	Hourly
Air Temperature	°C	Air temperature	Daily min/max/avg
Wind Speed	km/h	Wind speed	Daily min/max/avg
Rainfall	mm	Precipitation (not including snow)	Real daily value

Table 1: Data description summary

2.2 Data Pre-processing

Storm Surge Calculation

Storm surge is a key variable in flood and storm impact analysis, but it cannot be directly measured. Instead, it is derived using the following formula[11]:

$$\text{Storm Surge} = \text{Observed Water Level} - \text{Fitted Tide}$$

As mentioned previously, the observed water levels data for Halifax and Yarmouth are obtained from archived records provided by the Government of Canada.

To estimate the fitted tide component, the `ftide()` function from the `TideHarmonics` package in `R` is used. This function models tidal patterns based on harmonic constituents and requires three main inputs:

- `x`: A numeric vector object containing water level measurements, where missing values are allowed
- `dto`: A date-time vector (of class `POSIXct`) of the same length as `x`, where missing values are not permitted.
- `hcn`: A vector of harmonic constituent names. In this analysis, we use the four major constituents, `hc4`.

By applying this `ftide()` using the observed water level data, we generate fitted tide values, which are then subtracted from the observed levels to compute the storm surge. This approach enables accurate estimation of storm surge at both locations, based on tidal patterns and observed water level data. It is important to note that this stage of the pre-processing was completed as part of a pre-existing project, and the corresponding code has not been included in this project. Also, there are some references to water surge level in the analysis displaying later, that is equal to storm surge.

[11]Stephenson (2017)

Missingness Interpolation

In this study, linear interpolation was used to fill in missing values for the covariates. This method estimates missing values between two known data points by assuming a linear change between them. Suppose there are two known data points (t_1, y_1) and (t_2, y_2), and we aim to estimate the value at some point t where $t_1 < t < t_2$. The interpolated value is given by

$$y(t) = y_1 + \frac{y_2 - y_1}{t_2 - t_1} \times (t - t_1)$$

A cutoff of 15 days was applied, meaning that gaps of 15 days or fewer were interpolated using this method, while longer gaps were left untreated. However, linear interpolation assumes a constant rate of change between data points, which may oversimplify real-world variability, potentially smoothing over short-term fluctuations and underestimating uncertainty in the data.

In this analysis, daily average values of storm surge levels are computed and used alongside the corresponding daily average values of wind speed, rain, and temperature. After preprocessing, the final dataset includes 18,569 observations for Halifax and 19,582 for Yarmouth. The daily storm surge level and its associated covariates for Halifax are presented in Figure 1 below. From the plots, it appears that storm surge level and wind speed exhibit a similar pattern, especially when compared to the other two variables. The corresponding visualization for Yarmouth is provided in Figure 19 in the appendix.

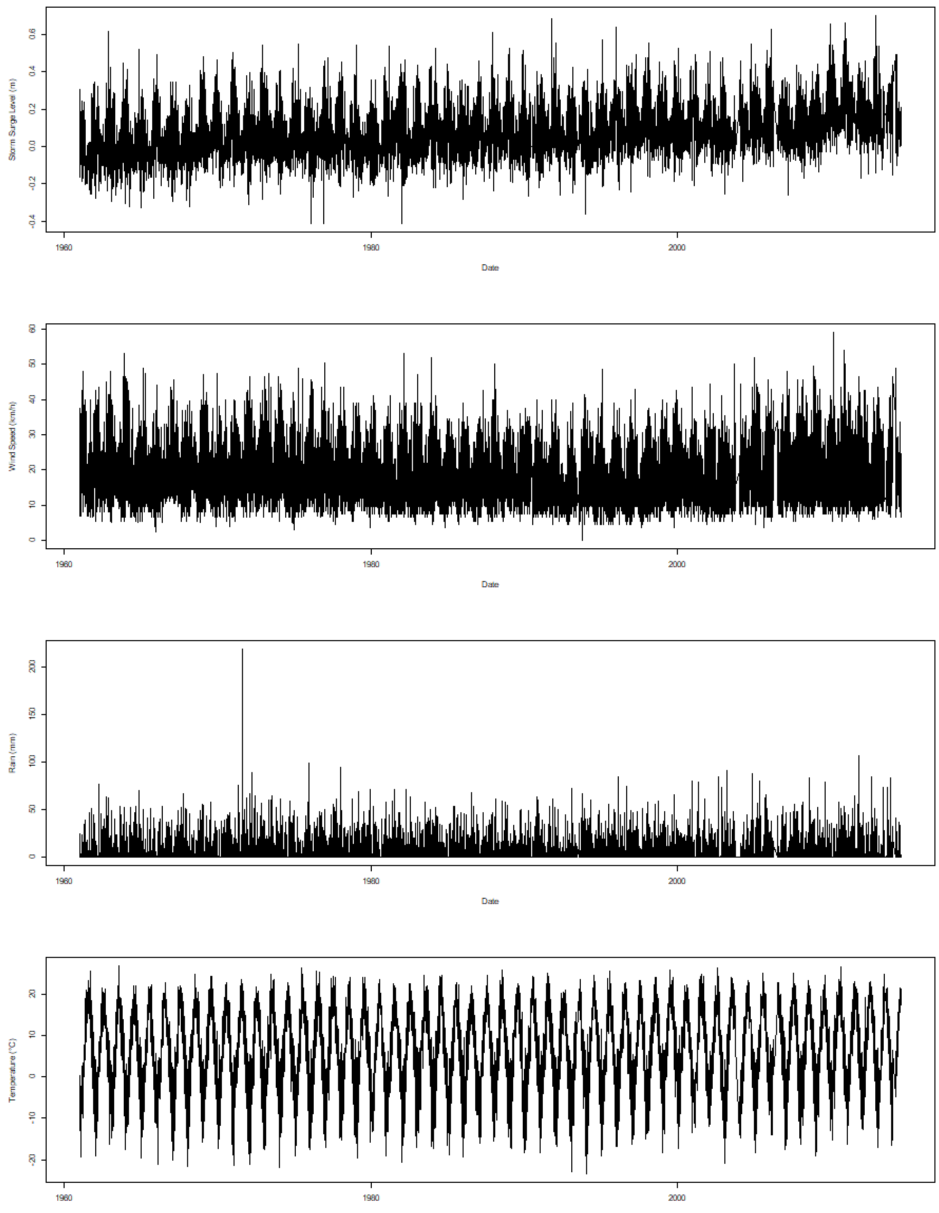

Figure 1: Daily time series of climatic factors in Halifax (storm surge, wind speed, rain, and temperature.)

3. Methods

3.1 Auto- and Cross-correlation (ACF and CCF)

The auto-correlation function is a scaled version of the auto-covariance, that is used to understand the dependence structure of the time series process. It measures how current values of the time series are related to past values across various lags. The auto-correlation of a general time series (X_t) at lag h is determined by

$$\rho_X(h) = \frac{\text{Cov}(X_{t+h}, X_t)}{\text{Var}(X_t)} = \frac{\gamma_X(h)}{\gamma_X(0)}$$

where $\gamma_X(h)$ is the auto-covariance function of the time series at lag h.

The cross-correlation is used to evaluate the relationship between two time series at different lags. For two general time series (X_t) and (Y_t), the cross-correlation at lag h is defined as

$$\rho_{XY}(h) = \frac{\text{Cov}(X_{t+h}, Y_t)}{\sqrt{\text{Var}(X_t)\text{Var}(Y_t)}} = \frac{\gamma_{XY}(h)}{\sqrt{\gamma_X(0)\gamma_Y(0)}}$$

where $\gamma_{XY}(h)$ is the cross-covariance function of two time series at lag h.

In `R`, functions `acf()`, and `ccf()` in package `stats` are used to compute and visualize the autocorrelation of a single time series and cross-correlation between two time series, respectively.

3.2 Generalized Additive Mixed Models (GAMMs)

Generalized additive mixed effect models (GAMMs) are statistical models that combine the flexibility of Generalized Additive Models (GAMs) with the ability to account for random effects in mixed-effect models. Specifically, GAMMs capture non-linear relationships between predictors and the response variable by applying smooth functions to each predictor. Moreover, it also incorporates random effects to account for group or cluster-level variability in the data. The model is typically expressed as:

$$y_i = f_Y(y_i; \mu_i, \theta)$$
$$g(\mu_i) = X\beta + Zw + \sum_{i=k}^{K} f_i(x_i)$$
$$w \sim N(0, \sigma^2)$$

where f_Y is the distribution of the response variable, g is the link function, f_i is the smooth function, $X\beta$ is the fixed effects, Zw is the random effects.

In addition, GAMMs assume smooth, additive relationships between predictors and the response, with random effects accounting for grouped or correlated data. In this study, both residuals and random effects are assumed to follow a normal distribution with constant variance, and the response variable is modeled as belonging to the exponential family. In `R`, the `mgcv` package provides the `gamm()` function, which fits GAMMs using the `nlme` package and allows for both non-linear predictor effects and random effects.

3.3 Vector Autogressive (VAR)

A VAR model is a more generalized version of the univariate autoregressive model[12]. It comprises one equation per variable in the system including a constant and lags of all of the variables in the system.

Consider $\mathbf{Y}_t$ be a $p \times 1$ vector of time series variables at time t. The VAR(k) model is determined by

$$\mathbf{Y}_t = \beta_0 + \Phi_1 \mathbf{Y}_{t-1} + \Phi_2 \mathbf{Y}_{t-2} + \cdots + \Phi_k \mathbf{Y}_{t-k} + \epsilon_t$$

where

$\mathbf{Y}_t = [y_{1t}, y_{2t}, \dots, y_{pt}]^T$ is the vector of p endogenous variables at time t,

β_0 is a $p \times 1$ vector of intercept terms,

Φ_i is a $p \times p$ matrix of autoregressive coefficients for lag $i = 1, \dots, k$,

ϵ_t is a $p \times 1$ vector of error terms assumed to be white noise: $\epsilon_t \sim N(0, \Sigma_\epsilon)$, where Σ_ϵ is the $p \times p$ covariance matrix.

In `R`, the `VAR()` function from the vars package can be used to fit a VAR(k) model for multivariate time series data (i.e., more than one time series). For a univariate time series, the `arima()` function from the `stats` package can be used to fit ARIMA models.

VAR model assumes that the lag structure in the VAR model is fixed and does not change over time. Moreover, the number of lags chosen can significantly affect the model's performance, which may cause overfitting or underfitting issues. To assess overfitting, rolling-origin cross-validation can be applied to a multivariate time series using a VAR model which can be visualized in Figure 2[13]. At each step, the VAR model is trained on an expanding window of the training data, and h-step ahead forecasts are produced, where $h = 1$ in the visualization. The forecast accuracy is calculated by averaging across test sets.

[12] Hyndman and Athanasopoulos (2018)
[13] Hyndman and Athanasopoulos (2018)

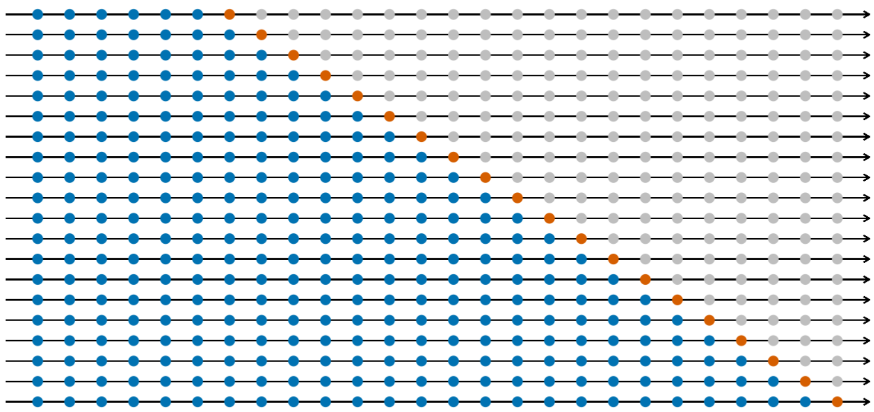

Figure 2: Visualization of the cross validation for time series with one step ahead forecast.

3.4 Granger Causality

Granger Causality is a statistical technique that determines whether one time series can predict another[14]. In a VAR(k) model with p variables, variable `x` is said to Granger-cause variable `y` if the past values of `x` provide statistically significant information about the future values of `y` that cannot be explained by the past values of `y` itself. The test determines whether the coefficients for the lagged values of `x` are jointly significant in the equation for `y`. If they are, we reject the null hypothesis, which holds that `x` does not cause `y`. However, it is important to note that Granger causality identifies predictive relationships rather than direct causal links.

Consider the system with only one target variable y_t with p explanatory variables $x_{1t}, x_{2t}, \dots, x_{pt}$ in VAR(k) model as follows:

$$y_t = \beta_0 + \sum_{i=1}^{k} \alpha_i y_{t-i} + \sum_{j=1}^{p} \sum_{i=1}^{k} \beta_{ji} x_{j,t-i} + \epsilon_t$$

where β_0 is an intercept, α_i are coefficients corresponding to the lagged of the target variable y_t, and β_{ji} are coefficients for the lags of explanatory variables x_j for $j = 1, \dots, p$.

The test for Granger causality in the above model involves in whether a specific variable x_j Granger-causes y_t, the null hypothesis is as follows:

$$H_0 : \beta_{j1} = \beta_{j2} = \dots = \beta_{jk} = 0$$

The rejection of the null hypothesis suggest that past values of x_j contain predictive information about y_t beyond what can be explained by the past values of y_t alone or we can

[14]Seth (2007)

say x_j Granger-causes y_t. In `R`, the function `vars::causality()` is commonly used to compute test statistics for Granger causality within the context of VAR(k). Additionally, `the lmtest::grangertest()` function can be used to perform Granger causality tests in a more general (bivariate) setting.

3.5 Extreme Value Analysis (EVA)

Given the focus on flooding incidents along the coast of Nova Scotia, it is important to identify and model the extreme storm surge levels associated with these events. Extreme Value Analysis (EVA) is a statistical framework designed to analyze, model, and predict the behavior of rare or extreme occurrences. Its goal is to estimate the probability of events that exceed all previously observed values. These extreme events are often characterized by their low probability of occurrence but high potential for severe consequences.

The EVA method employed in the second research question is the Block Maxima (BM) approach. This technique involves dividing the full observation period into non-overlapping intervals of equal length n, referred to as blocks. From each block, only the maximum observation—called the block maxima—is retained for analysis. Often, these blocks correspond to calendar years, resulting in annual maxima. These maximas are then modeled using the Generalized Extreme Value (GEV) distribution.

In this study, data from two locations—Halifax and Yarmouth—are considered. For each location j, let y_j denote the vector of annual maximas, which has length N_j, the number of years of data available at that site. Each entry y_{ji} in this vector represents the annual maximum for year i at location j. These annual maximas are assumed to follow the GEV distribution[15],

$$y_{ji} \sim GEV(\mu_j, \sigma_j, \xi_j), \text{for } i = 1, 2, \cdots, N_j, \text{ and } j = 1, 2.$$

The density function for y_{ji} is:

$$g(y \mid \mu_j, \sigma_j, \xi_j) = \frac{1}{\sigma_j}\left\{1 + \xi_j\left(\frac{y - \mu_j}{\sigma_j}\right)\right\} \exp\left\{-\left[1 + \xi_j\left(\frac{y - \mu_j}{\sigma_j}\right)\right]^{-\frac{1}{\xi_j}}\right\}.$$

where $\mu_j, \xi_j \in \mathbb{R}$, $\sigma_j > 0$, and,

$$y \in \left[\mu_j - \frac{\sigma_j}{\xi_j}, \infty\right] \text{ if } \xi_j > 0,$$
$$y \in (-\infty, \infty) \text{ if } \xi_j = 0,$$

[15] Coles (2001)

$$y \in \left[\infty, \mu_j + \frac{\sigma_j}{\xi_j}\right] \text{ if } \xi_j < 0.$$

These annual maximas are fitted to the GEV distribution using the `gev.fit()` function in `R`, which takes the vector of annual maximas as input. To assess the quality of the fit, the `gev.diag()` function can be used to produce diagnostic plots, helping to identify any notable deviations from the assumptions of the model, particularly normality.

Once the GEV model is fitted, maximum likelihood estimates (MLEs) for the three parameters—location (μ), scale (σ), and shape (ξ)—are obtained. These estimates are then used to calculate return levels corresponding to various return periods. A return level, denoted z_p, quantifies risk by representing the value expected to be exceeded on average once every $\frac{1}{p}$ years. For $0 < p < 1$, the MLE of the return level z_p is calculated as follows:

$$\hat{z}_p = \hat{\mu} - \frac{\hat{\sigma}}{\hat{\xi}}\left[1 - y_p^{-\hat{\xi}}\right], \ y_p = -\log(1-p)$$

.

For confidence interval calculations of these return levels, by the delta method we have,

$$\mathrm{Var}(\hat{z}_p) \approx \nabla z_p^T V \nabla z_p,$$

where V is the variance-covariance matrix of $(\hat{\mu}, \hat{\sigma}, \hat{\xi})$ and

$$\begin{aligned}\nabla z_p^T &= \left[\frac{\partial z_p}{\partial \mu}, \frac{\partial z_p}{\partial \sigma}, \frac{\partial z_p}{\partial \xi}\right] \\ &= \left[1, -\xi^{-1}(1 - y_p^{-\xi}), \sigma\xi^{-2}(1 - y_p^{-\xi}) - \sigma\xi^{-1}y_p^{-\xi}\log y_p\right]\end{aligned}$$

With that, we have $(1-\alpha) * 100\%$ confidence intervals constructed using,

$$CI(\hat{z}_p) = \hat{z}_p \pm z_{\alpha/2} \cdot \mathrm{SE}(\hat{z}_p), \text{where } \mathrm{SE}(\hat{z}_p) = \sqrt{\mathrm{Var}(\hat{z}_p)}$$

Return levels can also be estimated while incorporating covariates—such as wind speed—to potentially improve accuracy. In this approach, the model is modified so that the location parameter μ is expressed as a linear function of wind speed:

$$Z_i \sim GEV(\mu(x_i), \sigma, \xi)$$

where Z_i represents the annual maximum storm surge in year i, and x_i is the corresponding wind speed. Note that the scale (σ) and shape (ξ) parameters are assumed to be constant and independent of the covariates, and the location (μ) parameter is modeled as follows,

$$\mu(x) = \beta_0 + \beta_1 x$$

Although this covariate-based modeling was attempted using the `gev.fit()` function's covariate argument (as will be further discussed in the Analysis section), there are concerns that the implementation may have been incorrect or incomplete. Accurately incorporating covariates appears to require a more advanced treatment, which lies beyond the scope of this project.

4. Analysis

The Halifax location is used as the example in all the code chunks illustrated in this section, which are replicated for the Yarmouth location data.

4.1 Time Series Analysis and Covariates Interaction Exploration

Auto-correlation functions of each time series of climatic factors including storm surge, wind speed, temperature and rain in `R` are performed below.

```
# Storm surge
hf_acf_surge = acf(halifax$avg_water_surge, lag.max = 50, main = "Halifax-ACF-Storm Surge")

# Wind speed
hf_acf_wind = acf(halifax$avg_wind_speed, lag.max = 50, main = "Halifax-ACF-Wind Speed")

# Temperature
hf_acf_temp = acf(halifax$avg_temperature, lag.max = 50, main = "Halifax-ACF-Temperature")

# Rain
hf_acf_rain = acf(halifax$rain, lag.max = 50, main = "Halifax-ACF-Rain")
```

Cross-correlation functions of storm surge level and other climatic factors are given in the `R` code below.

```
# Storm surge and Wind speed
hf_ccf_surge_wind = ccf(halifax$avg_water_surge, halifax$avg_wind_speed, lag.max = 50,
    plot = F)

# Storm surge and Temperature
hf_ccf_surge_temp = ccf(halifax$avg_water_surge, halifax$avg_temperature, lag.max = 50,
    plot = F)
```

```r
# Storm surge and Rain
hf_ccf_surge_rain = ccf(halifax$avg_water_surge, halifax$rain, lag.max = 50, plot = F)
```

GAMMs incorporate smooth terms for temperature, wind speed, rainfall, and date, allowing for the capture of seasonal patterns and nonlinear trends in water surge dynamics. The model aims to investigate the relationship between water surge and climatic predictors, accounting for location as a random effect, which can be expressed by the following formula:

$$\begin{aligned}\text{storm\_surge}_i =& \beta_0 + \beta_1 \text{temp}_i + \beta_2 \text{wind\_speed}_i + \beta_3 \text{rain}_i + f_1(\text{temp}_i) \\ &+ f_2(\text{wind\_speed}_i) + f_3(\text{rain}_i) + f_4(\text{time}_i) + w_{g,i} + \epsilon_i\end{aligned}$$

where β_k is a coefficient, f_j is a smooth function, and $w_{g,i} \sim N(0, \sigma_g^2)$ is a random effect covariate accounting for variations in storm surge across different locations.

In `R`, the `gamm()` function is used as follows:

```r
library(nlme)
library(mgcv)

# Load data
new_df_linear_interpolation = read.csv("data/linear_interpolation_df.csv")
new_df_linear_interpolation$date_num = as.numeric(as.Date(new_df_linear_interpolation$date))

location_gamm <- gamm(
  avg_water_surge ~
    s(avg_temperature) +
    s(avg_wind_speed) +
    s(rain) +
    date_num,
  random = list(location = ~ 1),
  # Random effect for location
  data = new_df_linear_interpolation
)
```

Then the residuals of the smooth terms and random effects were checked as follows:

- First, we checked for auto-correlation of the residuals since the errors are assumed to be independent.

```
# Random Effect
acf(residuals(location_gamm$lme), main = "ACF of residuals of Random Effect part in GAMM")

# Smooth part
acf(residuals(location_gamm$gam), main = "ACF of residuals of Smooth part in GAMM")
```

- Secondly, a residual analysis was performed for both smooth and random effects, including a Residuals vs. Fitted plot and a Q-Q plot of the residuals.

```
# ---- Extract residuals of Random Effect ----
residuals_gamm_location_lme <- resid(location_gamm$lme)
residuals_pearson_location_lme <- residuals(location_gamm$lme, type = "pearson")
# Seem to scatter around 0
plot(fitted(location_gamm$lme), residuals_gamm_location_lme, xlab = "Fitted Values",
    ylab = "Residuals", main = "Residuals vs Fitted")
abline(h = 0, col = "red")
# QQ plot: not normal
qqnorm(residuals_gamm_location_lme, main = "QQ Plot of Residuals")
qqline(residuals_gamm_location_lme, col = "red")

# ---- Extract residuals of Smooth ----
residuals_gamm_location_gam <- resid(location_gamm$gam)
residuals_pearson_location_gam <- residuals(location_gamm$gam, type = "pearson")
# Seem to scatter around 0
plot(fitted(location_gamm$gam), residuals_gamm_location_gam, xlab = "Fitted Values",
    ylab = "Residuals", main = "Residuals vs Fitted")
abline(h = 0, col = "red")
# QQ plot: not normal
qqnorm(residuals_gamm_location_gam, main = "QQ Plot of Residuals")
qqline(residuals_gamm_location_gam, col = "red")
```

Since the residual analysis indicates a dependence structure and violations of model assumptions, this motivates the use of alternative approaches, such as Vector Autoregression (VAR) models to account for the temporal structure.

Vector Autoregression (VAR): The following model represents storm surge as a function of its own past values and the lagged values of three covariates including temperature, wind speed, and rainfall over 7 previous time steps:

$$\text{storm\_surge}_t = \sum_{i=1}^{7}(\beta_{1,i}\text{storm\_surge}_{t-i} + \beta_{2,i}\text{temp}_{t-i} + \beta_{3,i}\text{wind\_speed}_{t-i} + \beta_{4,i}\text{rain}_{t-i}) + \epsilon_t$$

where $\epsilon \sim N(0, \sigma^2)$.

Before fitting the VAR model, the Augmented Dickey-Fuller (ADF) test was performed to check the stationarity of the time series data.

```r
library(vars)
library(tseries)
library(forecast)

halifax_var <- new_halifax_df_interpolation[, c("avg_water_surge", "avg_temperature",
    "avg_wind_speed", "rain")]

# ADF test for each time series
adf_test <- apply(halifax_var, 2, adf.test)
adf_test
```

Then, the `VAR()` model can be applied as follows:

```r
halifax_var_model <- VAR(halifax_var, p = 7, type = "none")

# Model evaluation
summary(halifax_var_model$varresult$avg_water_surge)
AIC(halifax_var_model$varresult$avg_water_surge)

# Residual ACF plot
acf(residuals(halifax_var_model)[, 1], main = "VAR ACF Residuals Model 1 - Halifax")
```

No significant autocorrelation was observed in the residuals, suggesting that the model adequately captures the temporal dependence. However, to address potential overfitting, time series cross-validation was applied. The model's performance was evaluated using the Root Mean Square Error (RMSE) from both the overall model fit and the average RMSE across expanding training windows. This was implemented using a custom function specifically written for this purpose.

```r
# Model RMSE
rmse_model = sqrt(mean(halifax_var_model$varresult$avg_water_surge$residuals^2))

# Cross-validation RMSE
cv_res_1 = tsCV_VAR(halifax_var, start = 18000, h = 1, p = 7, target_var = "avg_water_surge")
cv_res_1$rmse
```

To simplify the model and focus on the most influential predictors, we narrowed the variables to storm surge and wind speed at lag 1. The reduced model is given by:

$$\text{storm\_surge}_t = \beta_0 + \beta_1 \text{storm\_surge}_{t-1} + \beta_2 \text{wind\_speed}_{t-1} + \epsilon_t$$

```
halifax_var2 <- halifax_var[c("avg_water_surge", "avg_wind_speed")]
halifax_var_model2 <- VAR(halifax_var2, p = 1, type = "none")

# Model evaluation
summary(halifax_var_model2$varresult$avg_water_surge)
AIC(halifax_var_model2$varresult$avg_water_surge)

# Residual ACF plot
acf(halifax_var_model2$varresult$avg_water_surge$residuals, main = "VAR ACF Residuals Model 
```

Finally, to evaluate whether the storm surge level can be effectively modeled by its own previous value alone, we fitted an AR(1) model:

$$\text{storm\_surge}_t = \beta_0 + \beta_1 \text{storm\_surge}_{t-1} + \epsilon_t$$

```
halifax_var_model3 = arima(halifax_var$avg_water_surge, order = c(1, 0, 0))
summary(halifax_var_model3)

# --- Model Evaluation --- Get fitted values
fitted_values <- halifax_var$avg_water_surge - residuals(halifax_var_model3)

# R-squared
SSE <- sum(residuals(halifax_var_model3)^2)
SST <- sum((halifax_var$avg_water_surge - mean(halifax_var$avg_water_surge))^2)
r_squared <- 1 - SSE/SST

# AIC
AIC(halifax_var_model3)

# --- ACF of Residuals ---
acf(na.omit(halifax_var_model3$resid), main = "VAR ACF Residuals Model 3 - Halifax")
```

Granger Causality is employed to examine whether wind speed, temperature, and rainfall at lag one Granger-cause storm surge levels. The following `R` code demonstrates how to compute the test statistics for Granger causality using a VAR(1) model that includes wind speed, temperature, and rainfall as potential predictors of storm surge.

```r
# Wind speed, temperature and rain at lag 1 Granger-cause storm surge?
halifax_var_model <- VAR(halifax_var, p = 1, type = "none")
causality(halifax_var_model, c("avg_wind_speed", "avg_temperature", "rain"))$Granger

# Wind speed at lag 1 Granger-causes storm surge?
halifax_var_model_wind <- VAR(halifax_var[, c("avg_water_surge", "avg_wind_speed")],
    p = 1, type = "none")
causality(halifax_var_model_wind, "avg_wind_speed")$Granger

# Temperature at lag 1 Granger-causes storm surge?
halifax_var_model_temp <- VAR(halifax_var[, c("avg_water_surge", "avg_temperature")],
    p = 1, type = "none")
causality(halifax_var_model_temp, "avg_temperature")$Granger

# Rain at lag 1 Granger-causes storm surge?
halifax_var_model_rain <- VAR(halifax_var[, c("avg_water_surge", "rain")], p = 1,
    type = "none")
causality(halifax_var_model_rain, "rain")$Granger
```

Additionally, to assess the predictive relationship between storm surge and climatic factors, bivariate Granger causality tests were performed using `lmtest::grangertest()` with a lag of 1. Each test evaluates whether a covariate provides additional information about future storm surge beyond its past values.

```r
# Wind speed
grangertest(avg_water_surge ~ avg_wind_speed, order = 1, data = na.omit(halifax_var))

# Temperature
grangertest(avg_water_surge ~ avg_temperature, order = 1, data = na.omit(halifax_var))

# Rain
grangertest(avg_water_surge ~ rain, order = 1, data = na.omit(halifax_var))
```

4.2 Extreme Value Analysis

The Extreme Value Analysis (EVA) methodology previously described in the Methods section is applied to address the second research question, which aims to predict future flooding events through the estimation of return levels. These return levels are the way of quantifying the risk through EVA. The original goal was to enhance these predictions by incorporating covariates identified as most strongly correlated with storm surge levels in the analysis of the first research question.

To support this comparison, return levels were first estimated without the inclusion of covariates. These baseline results serve as a reference against which the covariate-based estimates can be evaluated.The following code example, based on the Halifax dataset, illustrates the process: extracting the annual maximas, fitting them to the GEV distribution, and computing the return levels along with their confidence intervals.

```
# Annual max storm surges extracted from the data.
annual.maximas.hx = full.max.data.halifax %>%
    mutate(year = year(date)) %>%
    group_by(year) %>%
    summarise(max_storm_surge = max(avg_water_surge, na.rm = TRUE), .groups = "drop")

# Required libraries.
library(ismev)
library(evd)

# Fitting the annual maximas using the GEV distribution.
gev.surge = gev.fit(annual.maximas.hx$max_storm_surge)
gev.diag(gev.surge)

# Extract the GEV parameters from the fitted model
mu = gev.surge$mle[1]  # Location parameter (mu)
sigma = gev.surge$mle[2]  # Scale parameter (sigma)
xi = gev.surge$mle[3]  # Shape parameter (xi)
# The variance covariance matrix.
V = gev.surge$cov

# Define the return periods representing years.
T_values = c(2, 5, 10, 20, 50, 100)
p = 1/T_values
# yp will be used in the return levels confidence interval calculations.
yp = -log(1 - p)

# Calculate return levels for each return period
return_levels = sapply(T_values, function(T) {
    # The return levels are essentially the quantiles of the GEV at the MLE
    # from the GEV fit with probabilities calculated for the different return
    # periods.
    return_level = qgev(1/T, loc = mu, scale = sigma, shape = xi, lower.tail = FALSE)
})
grad = matrix(0, nrow = length(p), ncol = 3)
var_return_levels = rep(0, length(p))
for (i in 1:length(p)) {
```

```
    # Compute Gradient Vector ( zp) of Return Levels
    grad[i, ] = c(1, -xi^(-1) * (1 - yp[i]^(-xi)), sigma * xi^(-2) * (1 - yp[i]^(-xi)) -
        sigma * xi^(-1) * yp[i]^(-xi) * log(yp[i]))
    # Compute Variance of Return Levels
    var_return_levels[i] = t(grad[i, ]) %*% V %*% grad[i, ]
}

# Compute Standard Errors of Return Levels
se_return_levels = sqrt(var_return_levels)

# 95% Confidence Interval
alpha = 0.05
z_value = qnorm(1 - alpha/2)
lower.bounds = return_levels - z_value * se_return_levels
upper.bounds = return_levels + z_value * se_return_levels
```

Following the initial analysis without covariates, an attempt was made to incorporate wind speed as a covariate, as it was identified in the first research question as having a significant relationship with storm surge. The code example below demonstrates how this was implemented using the Halifax dataset.

```
# Annual max wind speed extracted from the data.
annual.maximas.Halifax_ws <- full.max.data.Halifax %>%
    mutate(year = year(date)) %>%
    group_by(year) %>%
    summarise(max_wind_speed = max(avg_wind_speed, na.rm = TRUE), .groups = "drop")

# Load required libraries
library(ismev)
library(evd)

# Transform wind speed as data frame
max_wind_speed = data.frame(max_wind_speed = annual.maximas.Halifax_ws$max_wind_speed)

# Fitting the GEV model with a covariate.
gev.surge_wind = gev.fit(xdat = annual.maximas.Halifax_ss$max_storm_surge, ydat = max_wind_sp
    mul = 1)
gev.diag(gev.surge_wind)

# Extract the parameters from the fitted model:
mu_intercept = gev.surge_wind$mle[1]  # Intercept for mu
mu_slope = gev.surge_wind$mle[2]  # Effect of wind speed on mu
```

```r
sigma = gev.surge_wind$mle[3]  # Scale parameter (must be > 0)
xi = gev.surge_wind$mle[4]  # Shape parameter
V = gev.surge_wind$cov  # Covariance matrix for parameters

# Choose a reference value for the covariate (e.g., mean wind speed)
mean_wind = mean(annual.maximas.Halifax_ws$max_wind_speed, na.rm = TRUE)

# Compute the location parameter mu for the reference wind speed
mu = mu_intercept + mu_slope * mean_wind

# Define the return periods (in years)
T_values = c(2, 5, 10, 20, 50, 100)
p = 1/T_values
yp = -log(1 - p)

# Calculate return levels for each return period using the qgev function.
return_levels = sapply(T_values, function(T) {
    qgev(1/T, loc = mu, scale = sigma, shape = xi, lower.tail = FALSE)
})

# Calculate the variance of the return levels via the delta method.  Since we
# now have 4 parameters, we define a gradient vector of length 4.
grad = matrix(0, nrow = length(p), ncol = 4)
var_return_levels = rep(0, length(p))

for (i in 1:length(p)) {
    # Here, we assume the derivative with respect to mu_slope is equal to the
    # reference covariate value (mean_wind).
    grad[i, ] = c(1, mean_wind, -xi^(-1) * (1 - yp[i]^(-xi)), sigma * xi^(-2) * (1 -
        yp[i]^(-xi)) - sigma * xi^(-1) * yp[i]^(-xi) * log(yp[i]))
    # Compute variance using the delta method
    var_return_levels[i] = t(grad[i, ]) %*% V %*% grad[i, ]
}

# Compute Standard Errors for the return levels
se_return_levels = sqrt(var_return_levels)

# 95% Confidence Intervals for return levels
alpha = 0.05
z_value = qnorm(1 - alpha/2)
lower_bounds = return_levels - z_value * se_return_levels
upper_bounds = return_levels + z_value * se_return_levels
```

5. Results

5.1 ACF and CCF

The ACF plots for Halifax and Yarmouth (Figure 3 and Figure 4) provide insight into how each variable correlates with itself over time. For both locations, storm surge, wind speed, temperature, and rainfall time series are examined. As expected, the plots display similar general patterns across locations, indicating that these variables behave in comparable ways.

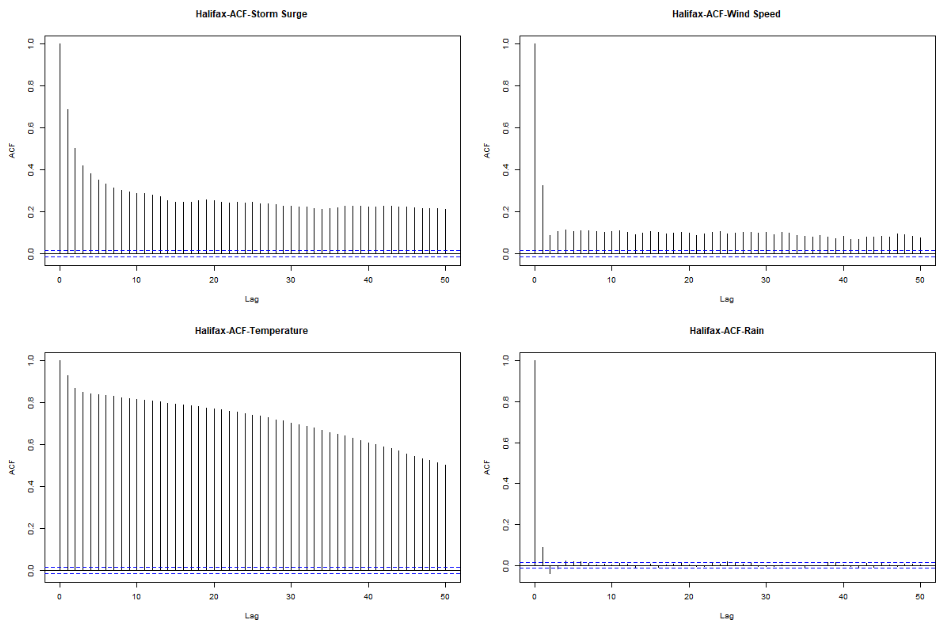

Figure 3: ACF plots of storm surge (top left), wind speed (top right), temperature (bottom left) and Rain (bottom right) in Halifax.

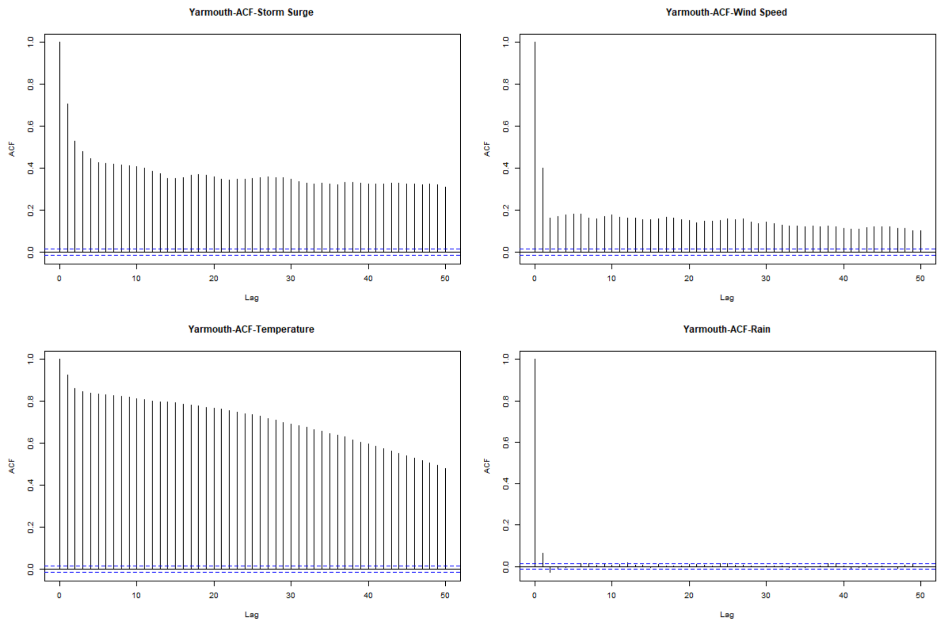

Figure 4: ACF plots of storm surge (top left), wind speed (top right), temperature (bottom left) and Rain (bottom right) in Yarmourth.

To be more specific:

- For storm surge (top-left plot), the ACF shows a strong temporal autocorrelation that gradually decreases but remains significant (above the blue confidence lines) even at 50 lags. This suggests that storm surge events exhibit persistent memory effects and follow cyclical patterns.
- For wind speed (top-right plot), the ACF reveals significant autocorrelation at lags 0 and 1, but the correlation rapidly declines thereafter. This suggests that wind events are relatively short-lived, with little long-term persistence.
- For temperature (bottom-left plot), the ACF exhibits the strongest and most persistent autocorrelation among all variables. The very gradual decline in correlation highlights a strong temporal persistence in temperature patterns, likely reflecting seasonal cycles and day-to-day stability.
- For rainfall (bottom-right plot), the ACF shows little to no significant autocorrelation beyond lag 0. This suggests that rainfall events are largely independent from one time period to the next, with minimal influence from past occurrences.

In summary, the ACF analysis highlights distinct temporal behaviors of covariates. Storm surges exhibit long-term dependencies and cyclical patterns, wind speed fluctuations are short-lived, temperature is highly persistent due to seasonal effects, and rainfall events occur independently with no significant memory. These differences reflect the underlying physical and meteorological drivers governing each variable.

The CCF plots reveal relationships between water surge and other variables, as shown in Figure 5 for Halifax and Figure 20 for Yarmouth Since the results for both locations are similar, only the analysis for Halifax is presented below:

- Storm Surge vs. Wind Speed (Top Plot): The CCF shows a positive correlation at lag 0, with a value of approximately 0.2, followed by a rapid decline in subsequent lags. This suggests wind events immediately impact water surge, with effects persisting through multiple time periods. The pattern indicates wind speed is a driver of water surge events.
- Storm Surge vs. Temperature (Middle Plot): A strong negative correlation is observed, peaking at lag 3 with a CCF value around -0.2. This suggests that increases in temperature are followed by decreases in storm surge levels after a few days. This inverse relationship is particularly pronounced for Halifax. That suggests lower temperatures are associated with higher storm surge events (possibly related to winter storm patterns).
- Storm Surge vs. Rainfall (Bottom Plot): The CCF reveals a short-term positive correlation, with the highest value (about 0.18) occurring at lag 0. The relationship weakens quickly over time, indicating that rainfall has a short-term impact on water surges but little long-term influence.

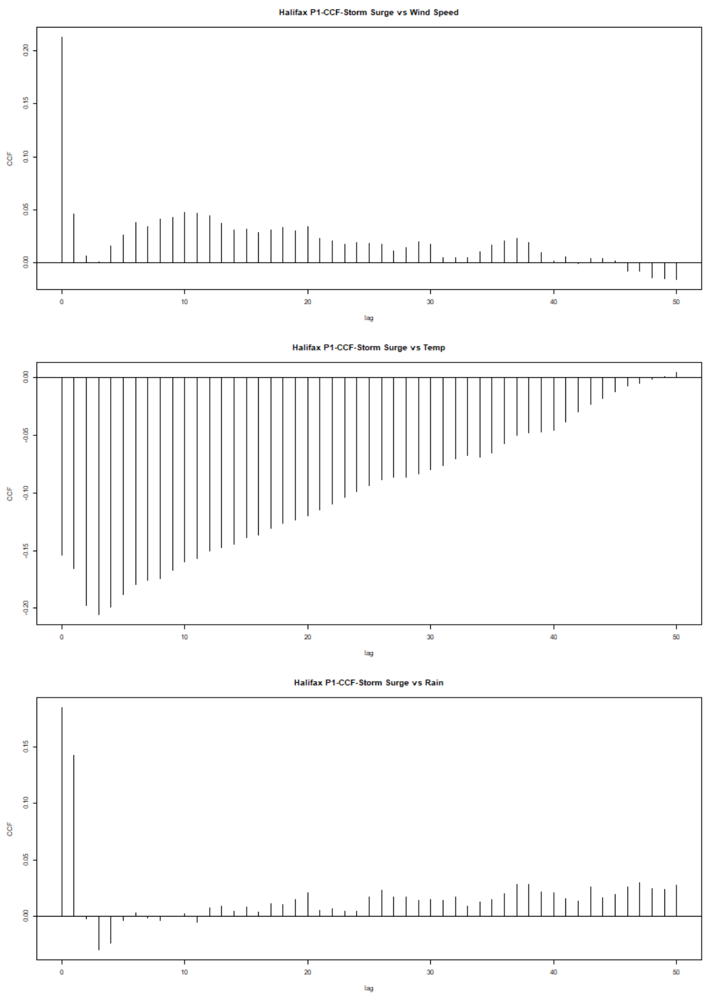

Figure 5: CCF plots of storm surge vs wind speed (top), storm surge vs temperature (middle), storm surge vs rain (bottom) in Halifax.

The patterns are generally similar between Halifax and Yarmouth, suggesting regional consistency in meteorological relationships. The temperature-surge negative correlation appears particularly strong in Halifax. Moreover, Halifax shows some interesting phase shifts in the wind-surge relationship that might indicate more complex coastal geography effects. The inverse relationship with temperature likely reflects seasonal storm dynamics. Overall, the findings suggest that storm surge events in Nova Scotia are primarily driven by wind conditions, with secondary influences from temperature variations and short-term rainfall.

5.2 GAMMs

The model outputs for the LME and GAM components are provided in Figure 6 and Figure 7, respectively. The LME results indicate a significant effect of location on storm surge, with wind speed and rainfall also having a statistically significant impact. Additionally, the GAM output suggests that the smooth functions of all three explanatory variables are significant in explaining the variability in storm surge.

```
> # Random Effect
> summary(location_gamm$lme)
Linear mixed-effects model fit by maximum likelihood
  Data: strip.offset(mf)
        AIC       BIC   logLik
  -64290.99 -64205.49 32155.49

Random effects:
 Formula: ~Xr - 1 | g
 Structure: pdIdnot
             Xr1       Xr2       Xr3       Xr4       Xr5       Xr6       Xr7       Xr8
StdDev: 0.2216509 0.2216509 0.2216509 0.2216509 0.2216509 0.2216509 0.2216509 0.2216509

 Formula: ~Xr.0 - 1 | g.0 %in% g
 Structure: pdIdnot
            Xr.01      Xr.02      Xr.03      Xr.04      Xr.05      Xr.06      Xr.07      Xr.08
StdDev: 0.03447626 0.03447626 0.03447626 0.03447626 0.03447626 0.03447626 0.03447626 0.03447626

 Formula: ~Xr.1 - 1 | g.1 %in% g.0 %in% g
 Structure: pdIdnot
           Xr.11    Xr.12    Xr.13    Xr.14    Xr.15    Xr.16    Xr.17    Xr.18
StdDev: 1.058269 1.058269 1.058269 1.058269 1.058269 1.058269 1.058269 1.058269

 Formula: ~1 | location %in% g.1 %in% g.0 %in% g
        (Intercept) Residual
StdDev:  0.04842095 0.104099

Fixed effects:  y ~ X - 1
                           Value  Std.Error    DF   t-value p-value
X(Intercept)           -0.03867090 0.03425206 38175  -1.12901  0.2589
Xdate_num               0.00000953 0.00000009 38175 106.65642  0.0000
Xs(avg_temperature)Fx1 -0.00507296 0.02135475 38175  -0.23756  0.8122
Xs(avg_wind_speed)Fx1   0.01414395 0.00614610 38175   2.30129  0.0214
Xs(rain)Fx1            -0.03667080 0.01125477 38175  -3.25825  0.0011
 Correlation:
                       X(Int) Xdt_nm X(_)F1 X(__)F
Xdate_num              -0.020
Xs(avg_temperature)Fx1  0.000  0.000
Xs(avg_wind_speed)Fx1   0.000  0.003 -0.001
Xs(rain)Fx1             0.000 -0.002 -0.001  0.000

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.32793139 -0.62616310 -0.03361181  0.58269124  5.40284984

Number of Observations: 38181
Number of Groups:
                              g                        g.0 %in% g               g.1 %in% g.0 %in% g
                              1                                 1                                 1
location %in% g.1 %in% g.0 %in% g
                              2
```

Figure 6: GAMMs output of LME

```
> # Smooth Term
> summary(location_gamm$gam)

Family: gaussian
Link function: identity

Formula:
avg_water_surge ~ s(avg_temperature) + s(avg_wind_speed) + s(rain) +
    date_num

Parametric coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -3.867e-02  3.425e-02  -1.129    0.259
date_num     9.527e-06  8.932e-08 106.661   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Approximate significance of smooth terms:
                     edf Ref.df     F p-value
s(avg_temperature) 8.223  8.223 145.6  <2e-16 ***
s(avg_wind_speed)  3.711  3.711 119.8  <2e-16 ***
s(rain)            7.989  7.989 274.4  <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

R-sq.(adj) =  0.207
  Scale est. = 0.010837  n = 38181
```

Figure 7: GAMMs output of GAM

Figure 8 shows the ACF of residuals, revealing a strong temporal structure in the data. The random effect component exhibits high autocorrelation at lag 1 (around 0.7), which gradually decreases but remains above the significance threshold beyond 45 lags. This indicates temporal dependencies in the data, violating the assumption of residual independence. Similarly, the smooth component shows a comparable autocorrelation at lag 1 with a more persistent decline, further suggesting that the independence assumption is not fully satisfied.

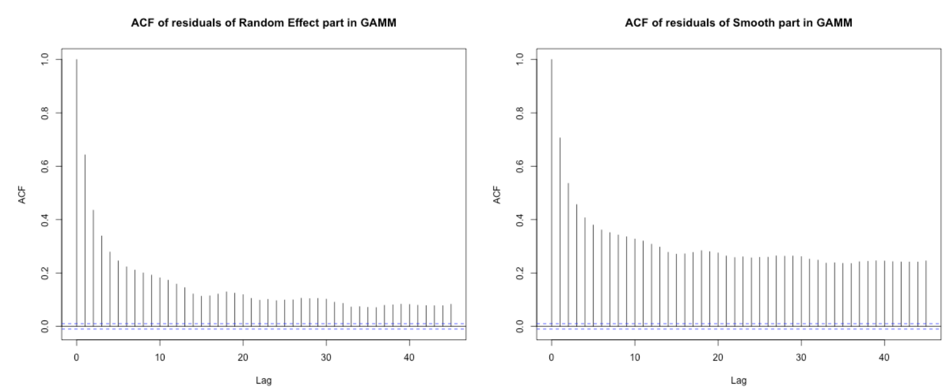

Figure 8: ACF plots of residuals in both Smooth and Random Effects in GAMMs

Residual diagnostic plots provide further insight into the model's performance:

- The residuals versus fitted values plot for the GAM component (Figure 9) shows a fairly uniform spread around zero, while the quantile-quantile (QQ) plot indicates deviations from normality.

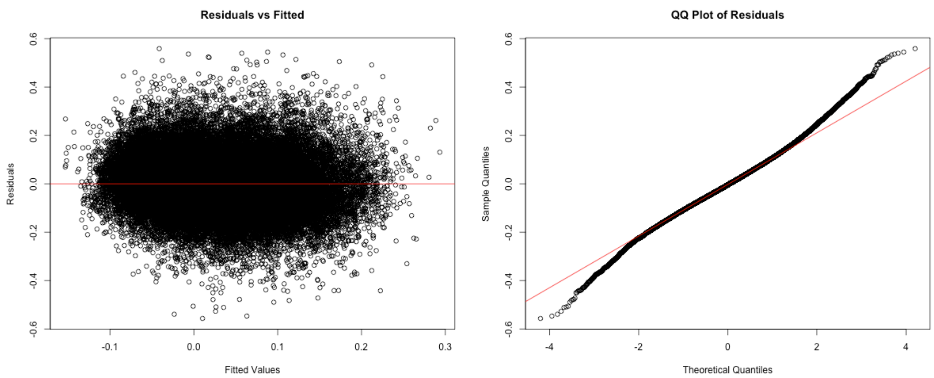

Figure 9: Residual diagnostic plots of GAM in GAMMs

- The residuals from the linear mixed effects (LME) component (Figure 10) exhibit a similar pattern, suggesting that the model does not fully capture the underlying structure of the data.

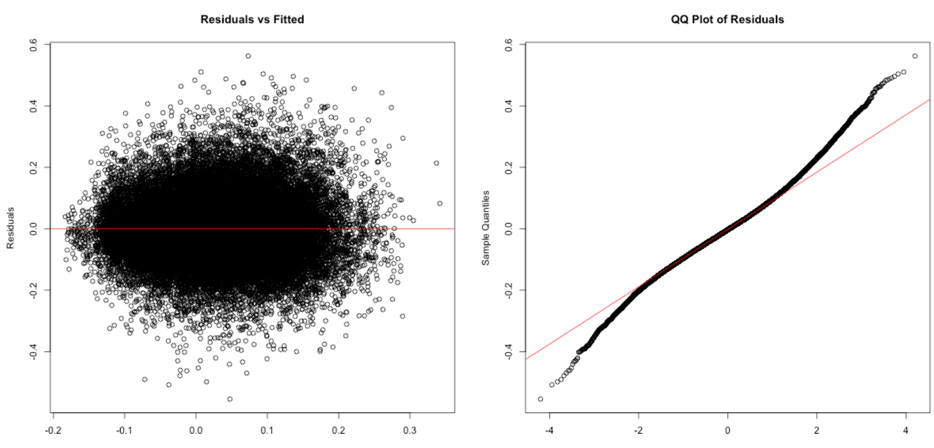

Figure 10: Residual diagnostic plots of LME in GAMMs

The model offers a strong foundation for understanding water surge dynamics, but adjustments to better account for temporal dependencies could further enhance its predictive performance and overall robustness. The inclusion of an explicit autocorrelation structure, such as a first-order autoregressive (AR(1)) correlation term, could help in capturing the remaining dependencies. Additionally, incorporating more time-related predictors or interactions may enhance explanatory power which leads us to VAR model.

5.3 VAR and Granger Causality

The VAR models were applied to both locations, and since the results are similar, only the results for Halifax are presented. Before model estimation, the stationarity of all time series variables—average water surge, average temperature, average wind speed, and rainfall-was tested using the Augmented Dickey-Fuller (ADF) test. The results below indicated that all variables were stationary (`p-value = 0.01` for each), permitting the use of VAR modeling without the need for differencing.

```
$avg_water_surge

    Augmented Dickey-Fuller Test

data:  newX[, i]
Dickey-Fuller = -17.295, Lag order = 26, p-value = 0.01
alternative hypothesis: stationary


$avg_temperature

    Augmented Dickey-Fuller Test

data:  newX[, i]
Dickey-Fuller = -7.8782, Lag order = 26, p-value = 0.01
alternative hypothesis: stationary


$avg_wind_speed

    Augmented Dickey-Fuller Test

data:  newX[, i]
Dickey-Fuller = -15.026, Lag order = 26, p-value = 0.01
alternative hypothesis: stationary


$rain
```

```
    Augmented Dickey-Fuller Test

data:  newX[, i]
Dickey-Fuller = -24.35, Lag order = 26, p-value = 0.01
alternative hypothesis: stationary
```

The initial model used a VAR(7) specification, incorporating all climatic factors—average water surge, temperature, wind speed, and rainfall—using seven lags. The primary target variable in the analysis was water surge. The choice for the number of lags was based on the observation that increasing the number of lags beyond seven did not significantly affect the AIC value, leading to a more complicated model that would be harder to interpret. The output of the VAR(7) model for Halifax, with the first 10 coefficients ordered by increasing p-value, is presented below. All presented covariates are statistically significant at the 1% level, suggesting a strong influence of their respective lagged values on the current storm surge level.

```
                       Estimate   Std. Error    t value     Pr(>|t|)
avg_water_surge.l1  0.6152109145 7.921935e-03  77.659175 0.000000e+00
avg_wind_speed.l1  -0.0017477502 9.811705e-05 -17.812911 2.167159e-70
rain.l2            -0.0011377409 7.633787e-05 -14.904017 6.042093e-50
rain.l1             0.0009455424 7.581813e-05  12.471190 1.489265e-35
avg_temperature.l2 -0.0027492896 2.582686e-04 -10.645080 2.191011e-26
avg_temperature.l1  0.0018600344 1.977047e-04   9.408144 5.624041e-21
avg_wind_speed.l2   0.0009737867 1.062547e-04   9.164646 5.477979e-20
avg_water_surge.l7  0.0517848817 7.950486e-03   6.513423 7.534469e-11
avg_water_surge.l2  0.0589565765 9.611218e-03   6.134142 8.735622e-10
avg_wind_speed.l6   0.0005108945 1.061634e-04   4.812343 1.503458e-06
```

- This model explained approximately 63.1% of the variation in storm surge (`R-squared = 0.631`) and exhibited the AIC of -39,456.38.
- Among the most significant predictors, the strongest influence is observed for the storm surge level at lag 1, with an estimated coefficient of 0.6152. This indicates a strong positive relationship, suggesting that current storm surge levels are influenced by their previous past values.
- Wind speed has a negative effect at lag 1 (-0.00175) and a positive effect at lag 2 (0.000974) on storm surge. These smaller coefficients are due to wind speed being measured in km/h, which tends to have larger numerical values.
- Rainfall displays a negative effect at lag 2 (-1.14e-03), and a positive effect for lag 1 (9.46e-04). As with wind speed, the smaller coefficients are a result of rainfall being measured in mm, which also involves relatively high magnitudes.

- Temperature also shows a negative effect at lag 2 (-2.75e-03), and a positive effect for lag 1 (1.86-03). These coefficients are smaller than that of water surge at lag 1, likely due to temperature being measured in degrees Celsius, which has a larger numerical scale.
- Additionally, the autocorrelation function (ACF) in Figure 11 indicates no significant autocorrelation, confirming that the model effectively captured temporal dependencies.

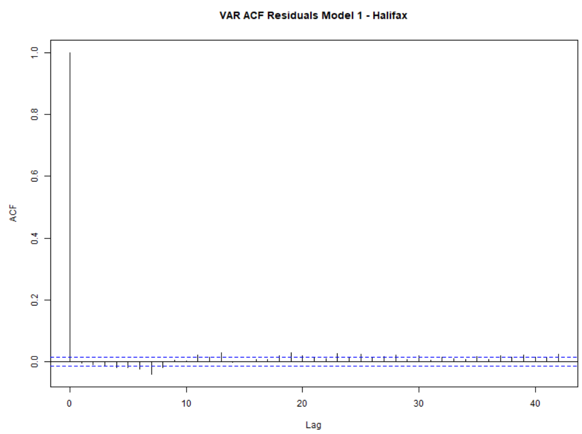

Figure 11: ACF plot of residuals of VAR(7) model for Halifax.

- Furthermore, time series cross-validation was used to assess potential overfitting. The model RMSE was 0.0839, and the average RMSE across expanding training windows was 0.0916, indicating no signs of overfitting.

A simplified model using VAR(1) incorporates only water surge and wind speed with a single lag is fitted to evaluate whether wind speed, as a significant covariate, can be used to forecast water surge one step ahead. The model having AIC of -37,594.7, which is larger than the VAR(7) model, with the output shown as follows:

```
Call:
lm(formula = y ~ -1 + ., data = datamat)

Residuals:
     Min       1Q   Median       3Q      Max
-0.62769 -0.04120  0.00193  0.05088  0.61280

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
avg_water_surge.l1 7.101e-01  5.539e-03  128.19   <2e-16 ***
avg_wind_speed.l1  7.290e-04  3.994e-05   18.25   <2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08791 on 18566 degrees of freedom
Multiple R-squared:  0.5906,    Adjusted R-squared:  0.5906
F-statistic: 1.339e+04 on 2 and 18566 DF,  p-value: < 2.2e-16
```

The VAR(1) model accounts for 59% of the variance in storm surge (`R-squared = 0.59`) and has a higher AIC value of -37,666.21, indicating a relatively poorer fit compared to the more complex VAR(7) model. Despite this, both included predictors—storm surge and wind speed—remain highly significant (`p < 2e-16`), with wind speed showing a consistent positive effect (with the estimated coefficient of 7.290e-04) on storm surge levels. The residual autocorrelation, as shown in Figure 12, appears acceptable, suggesting that the simplified model still captures key temporal dynamics.

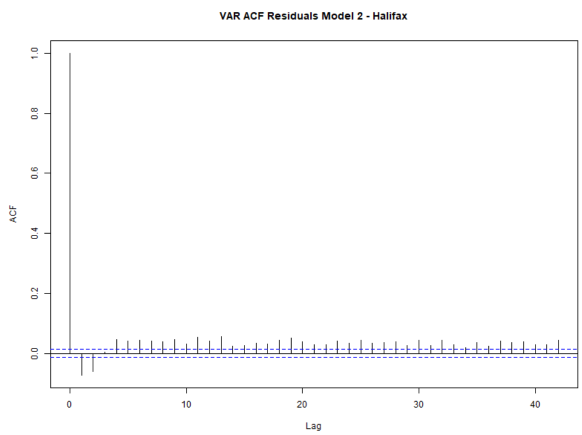

Figure 12: ACF plot of residuals of VAR(1) model of water surge and wind speed in Halifax.

To further isolate the influence of past water surge levels, the first order autoregressive model was estimated - AR(1). The result is shown as follows:

```
Call:
arima(x = halifax_var$avg_water_surge, order = c(1, 0, 0))

Coefficients:
         ar1  intercept
      0.6860     0.0684
s.e.  0.0053     0.0020
```

```
sigma^2 estimated as 0.00752:  log likelihood = 19054.9,  aic = -38103.81

Training set error measures:
                       ME       RMSE        MAE      MPE     MAPE      MASE
Training set 4.669154e-07 0.08671545 0.06300452 173.0141 424.5072 0.9329517
                    ACF1
Training set -0.04108029

[1] "R-squared: 0.4706"
```

The AR(1) model, focusing exclusively on the temporal dependence of water surge, has an R-squared of 0.4706, explaining 47% of the variance. The AIC value was -38,103.81, positioning it between the simplified VAR(1) model and the full VAR(7) model. The autoregressive coefficient (0.6860) indicates a strong temporal dependence, reinforcing the role of past water surge levels as the primary determinant of future surges. Moreover, the residual autocorrelation drops off quickly, suggesting the model adequately captures the underlying temporal structure.

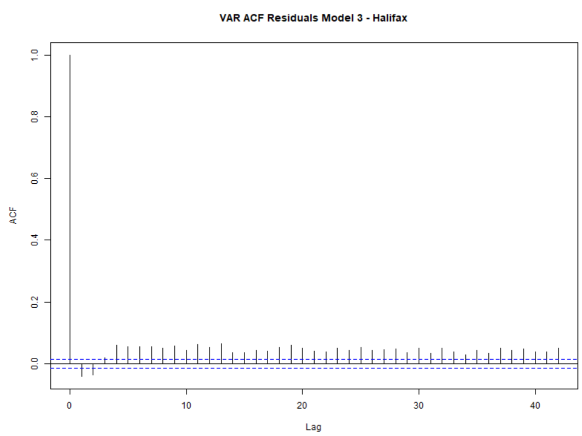

Figure 13: ACF plot of residuals of AR(1) model of storm surge level in Halifax.

The results highlight key insights into the dynamics of water surge variation. Water surge exhibits strong autocorrelation, with previous surge levels serving as the most influential predictor. The inclusion of climatic factors significantly improves predictive power, increasing explained variance from 47% (AR model) to 62% (full VAR model). Wind speed exerts both immediate and delayed effects, displaying an initial negative influence followed by positive effects at later lags, suggesting a complex interaction with water dynamics. Rainfall and temperature show a negative impact on storm surge at lag 2 and a positive impact at lag 1. Both effects are statistically significant at the 1% significance level, indicating that these climatic factors influence storm surge both in the short term (lag 1) and with a delayed effect (lag 2).

Overall, while the full VAR(7) model provides the best fit for accurately capturing storm surge dynamics, the VAR(1) model with wind speed represents a viable alternative for operational use, balancing predictive capability with model simplicity. These findings underscore the complex interplay between past water surge levels and climatic factors in shaping surge dynamics in Halifax, emphasizing the importance of multivariate approaches for improved forecasting.

Granger Causality:

Granger causality tests were conducted to examine the predictive power of individual climatic variables on storm surge. Each test compared a restricted model (including only lagged storm surge) against a full model that also included a lagged climatic predictor. To be specific:

- All three covariates at lag 1 combined: When all three climatic variables are jointly tested at lag 1, the null hypothesis that they do not Granger-cause storm surge is strongly rejected (`F = 118.31, p < 2.2e-16`). This result indicates that the lagged climatic factors provide significant predictive information for storm surge dynamics.
- Wind Speed at lag 1 significantly improves model performance, showing the strongest Granger causality effect (`F-Test = 333.05, p < 2.2e-16`). This suggests wind speed at lag 1 is the most dominant short-term predictor of storm surge variation.
- Rainfall at lag 1 also significantly contributes to the predictive model for storm surge dynamics (`F-Test = 89.911, p < 2.2e-16`), suggesting a causal relationship.
- Temperature although weaker than the other variables, still shows a significant effect at lag 1 (`F-Test = 41.206, p = 1.387e-10`), supporting its short-term predictive relevance.

```
[1] "All covariates"


    Granger causality H0: avg_temperature avg_wind_speed rain do not
    Granger-cause avg_water_surge

data:  VAR object halifax_var_model
F-Test = 118.31, df1 = 3, df2 = 74256, p-value < 2.2e-16

[1] "Wind speed"


    Granger causality H0: avg_wind_speed do not Granger-cause
    avg_water_surge

data:  VAR object halifax_var_model_wind
F-Test = 333.05, df1 = 1, df2 = 37132, p-value < 2.2e-16
```

```
[1] "Temperature"


    Granger causality H0: avg_temperature do not Granger-cause
    avg_water_surge

data:  VAR object halifax_var_model_temp
F-Test = 41.206, df1 = 1, df2 = 37132, p-value = 1.387e-10

[1] "Rain"


    Granger causality H0: rain do not Granger-cause avg_water_surge

data:  VAR object halifax_var_model_rain
F-Test = 89.911, df1 = 1, df2 = 37132, p-value < 2.2e-16
```

5.4 EVA

The extraction of annual maxima for each location enables the visualization of extreme values over time. As outlined in the Analysis section, these annual maxima are initially modeled using the Generalized Extreme Value (GEV) distribution without incorporating the wind speed covariate. Figures Figure 24 and Figure 25 display the annual maxima over time and the corresponding GEV diagnostic plots, respectively.

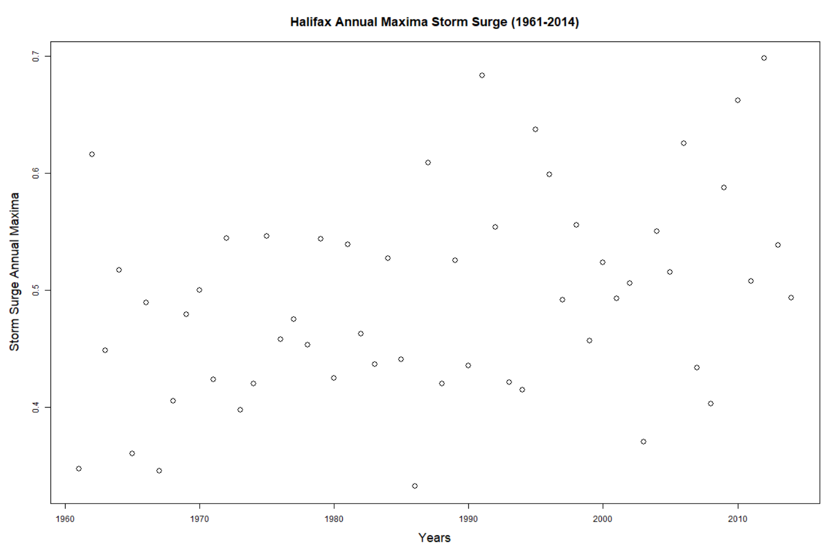

Figure 14: Halifax Annual Maximas Plotted against time

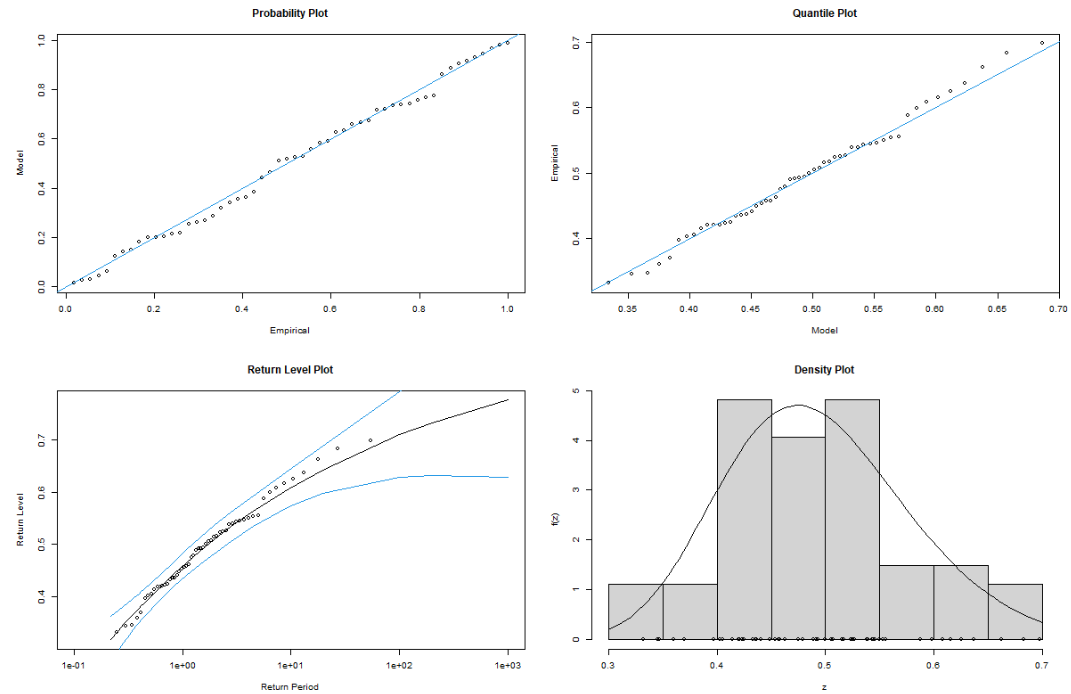

Figure 15: Halifax Diagnostics plots from the GEV fit - Non Covariate Case

From these figures, two key observations can be made:

- The annual maximas exhibit a subtle positive linear trend over time. This suggests the presence of non-stationarity inherent in the data. Although this non-stationarity likely contributes to some uncertainty in the model results, addressing it falls outside the scope of this project.
- The GEV diagnostic plots do not indicate any significant deviations, suggesting that the GEV model provides an adequate fit to the data.

The maximum likelihood estimates (MLEs) of the three GEV parameters obtained from this fit are summarized in the following table:

Parameter	**MLE Estimate**
Location ($\hat{\mu}$)	0.46
Scale ($\hat{\sigma}$)	0.08
Shape ($\hat{\xi}$)	-0.18

Table 2: Maximum Likelihood Estimates (MLEs) for the GEV parameters.

These parameters have some accompanying interpretations:

- $\hat{\mu} = 0.46$ is the baseline strom surge level expected annually.
- $\hat{\sigma} = 0.08$ is the level of variability between the annual maximas.
- $\hat{\xi} = -0.08 < 0$ is an indication that there is a bounded upper tail which is a physical cap on how high these storm surge levels can get.

Using the obtained MLE estimates, return level plots corresponding to return periods of 2, 5, 10, 20, 50, and 100 years can be constructed. Figure Figure 16 presents the return level plot for storm surges in Halifax, including confidence intervals for each return level estimate.

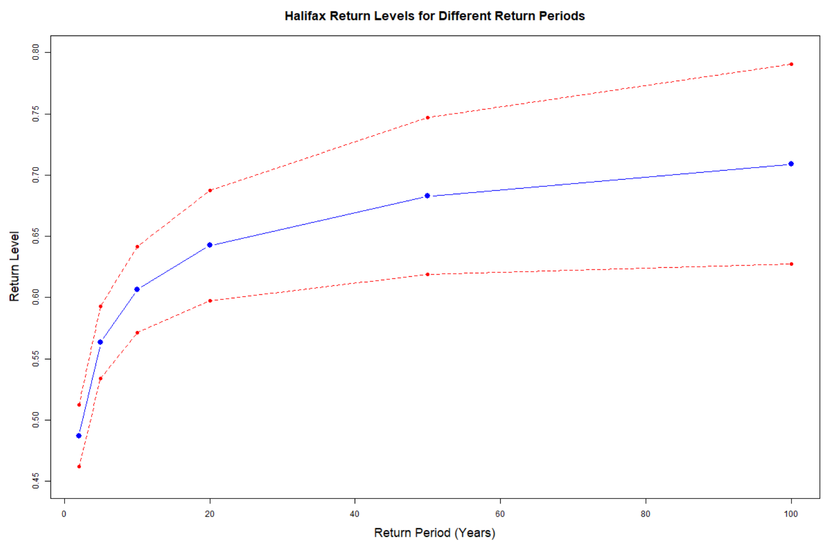

Figure 16: Halifax Return Level Plot - Non Covariate Case

Table~3 below presents the return level estimates in greater detail, complementing the information shown in Figure 16.

Return Period (years)	2	5	10	20	50	100
Storm Surge Level (m)	0.49	0.56	0.61	0.64	0.68	0.71

Table 3: Halifax Storm surge levels (in meters) corresponding to various return periods.

Looking at the overall pattern, both locations display higher storm surge return level for the longer return periods, following a logarithmic curve that flattens at higher return periods. This is typical in extreme value analysis with the difference between a 50-year and 100-year event is less dramatic than between a 2-year and 10-year event. Comparing both location, Halifax consistently shows higher return levels (approximately 0.08-0.10m higher) compared to Yarmouth across all return periods. Halifax's 100-year return level reaches about 0.71m while Yarmouth's reaches about 0.62m. See Figure 24, Figure 25, Figure 16, and Table~5 for references to the duplicated process for the Yarmouth location in the appendix.

Now, with respect to the incorporation of the covariates, the storm surge return levels were computed as explained in Analysis section. Figure 17 displays a visual of the wind speed data, while Figure 18 is the visual plot of the resulting return levels. See Figure 27, Figure 28, Figure 16, and Table~6 for references to the duplicated process for the Yarmouth location in the appendix.

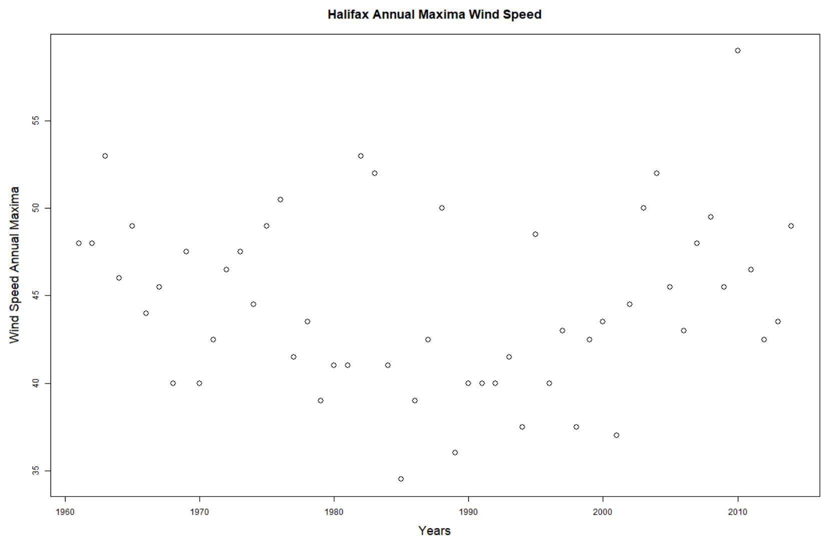

Figure 17: Halifax Wind Speed Plot

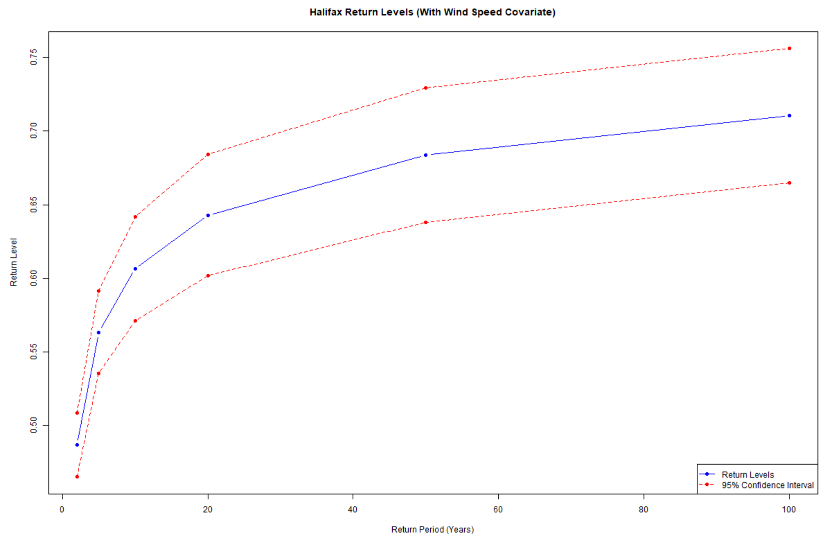

Figure 18: Halifax Return Level Plot - Covariate Case

Table~4 below presents the return level estimates in greater detail, complementing the information shown in Figure 18.

Return Period (years)	2	5	10	20	50	100
Storm Surge Level (m)	0.49	0.56	0.60	0.64	0.68	0.71

Table 4: Halifax Storm surge levels (in meters) corresponding to various return periods - Covariate Case.

It is clear that the covariate analysis produced remarkably similar results to the initial analysis, with differences so minimal they are indistinguishable at the hundredth decimal place displayed in the Table~3 and Table~4. For Halifax, differences between initial and covariate models are extremely small (0.0001-0.0003m). For Yarmouth, differences are slightly larger but still minor (0.0049-0.0116m). Both graphs show upper and lower confidence bounds (red dotted lines). The confidence interval widens as the return period increases, reflecting greater uncertainty in predicting more extreme events.

The minimal difference between the initial and covariate models suggests that wind speed does not significantly enhance the prediction of storm surges beyond what is already captured by the initial EVA model with no covariate. This may indicate that wind speed is implicitly reflected in the storm surge data, that the relationship between wind speed and storm surge in these locations is not strong enought to improve predictions, or that other environmental factors play a more critical role in driving extreme storm surge events. More plausibly, the similarity in results may stem from a suspected incorrect implementation of the wind speed covariate in the model, as mentioned in an earlier section. Given this uncertainty, future work could explore orecasting of storm surge levels using wind speed as a covariate through alternative methods—such as Long Short-Term Memory (LSTM) networks—which may offer smoother implementation and improved predictive performance.

6. Conclusion and Discussion

6.1 Conclusion

In conclusion for question one, across multiple analytical techniques—including CCF, GAMMs, VAR models, and Granger causality tests—there is consistent evidence that, beyond the strong temporal dependence of storm surge itself, wind speed significantly contributes to variations in surge levels in both Halifax and Yarmouth. Temperature shows a generally negative relationship with surge, likely linked to seasonal patterns, while rainfall exerts a more limited, short-term effect.The alignment of findings across locations and modeling frameworks reinforces the robustness of these relationships and highlights the critical role of wind in understanding storm surge dynamics in Nova Scotia.

As part of the continued analysis addressing question two, the Extreme Value Analysis (EVA) suggests that Halifax generally experiences slightly higher storm surge extremes than Yarmouth. However, incorporating wind speed as a covariate resulted in minimal to no improvement in predictive performance. This indicates that either a more accurate implementation of the covariate within the EVA framework, or the use of an alternative modeling approach, may be more effective for forecasting extreme events—an idea that will be explored further in the following section.

6.2 Discussion

Initially, this study aimed to analyze storm surge dynamics across multiple locations in Nova Scotia. However, due to limited data availability, only two locations were included. Missing values in both storm surge and covariate data posed a challenge. Accounting for the missingnesss in our storm surge variable would require other methods such as the Bayesian Hierarchial Model, which is beyond the scope of this project. While linear interpolation was applied to impute missing covariate values, it does not account for uncertainty, which may introduce bias. To preserve data integrity, we only used covariate values corresponding to available surge data. Additionally, we assumed equal time spacing between observations, which may affect estimation accuracy.

For the first question, regarding the interaction between covariates and storm surge levels using the VAR model, one limitation is the assumption of a fixed lag structure over time. This may not hold over long periods, potentially affecting the model's validity. Additionally, the VAR model's assumption of stationarity may not fully capture the evolving nature of the underlying climatic processes. Moreover, although Granger causality tests indicate statistical dependence, they do not imply true causality and are limited to detecting linear relationships.

For the second question, the aim was to use question one results to improve storm surge return level estimates. Initially only storm surge data was used for the predictions with the intention to make comparison for when there is incorporation of the correlated covariate. Results were not as expected, which raised several discussion points regarding the analysis:

- The highest correlated and most statistically significant covariate was selected for the model. However, adding this covariate did not provide additional useful information. Perhaps experimenting with less correlated variables could lead to more meaningful improvements.
- Additionally, a flawed implementation could be the cause of this unsatisfactory results. Even when applying the formula as suggested in description of the used R function, the literature indicates that incorporating covariates requires a more complex approach than initially assumed. Either a more correct implementation can be completed or a different method such as Long Short-Term Memory (LSTM) models may serve as an alternative. Because LSTMs can capture long-term dependencies and may provide more effective predictions for extreme surge events.
- A critical aspect of the model, which was briefly indicated previously, is the existing non-stationarity as part of the nature of the data. Our response variable, storm surge, is influenced by sea level, which is known to be increasing over time. Consequently, storm surge is also expected to rise over time. Despite this and for simplicity, non-stationarity was not addressed as it is expected to be a much more complicated procedure.

Appendix

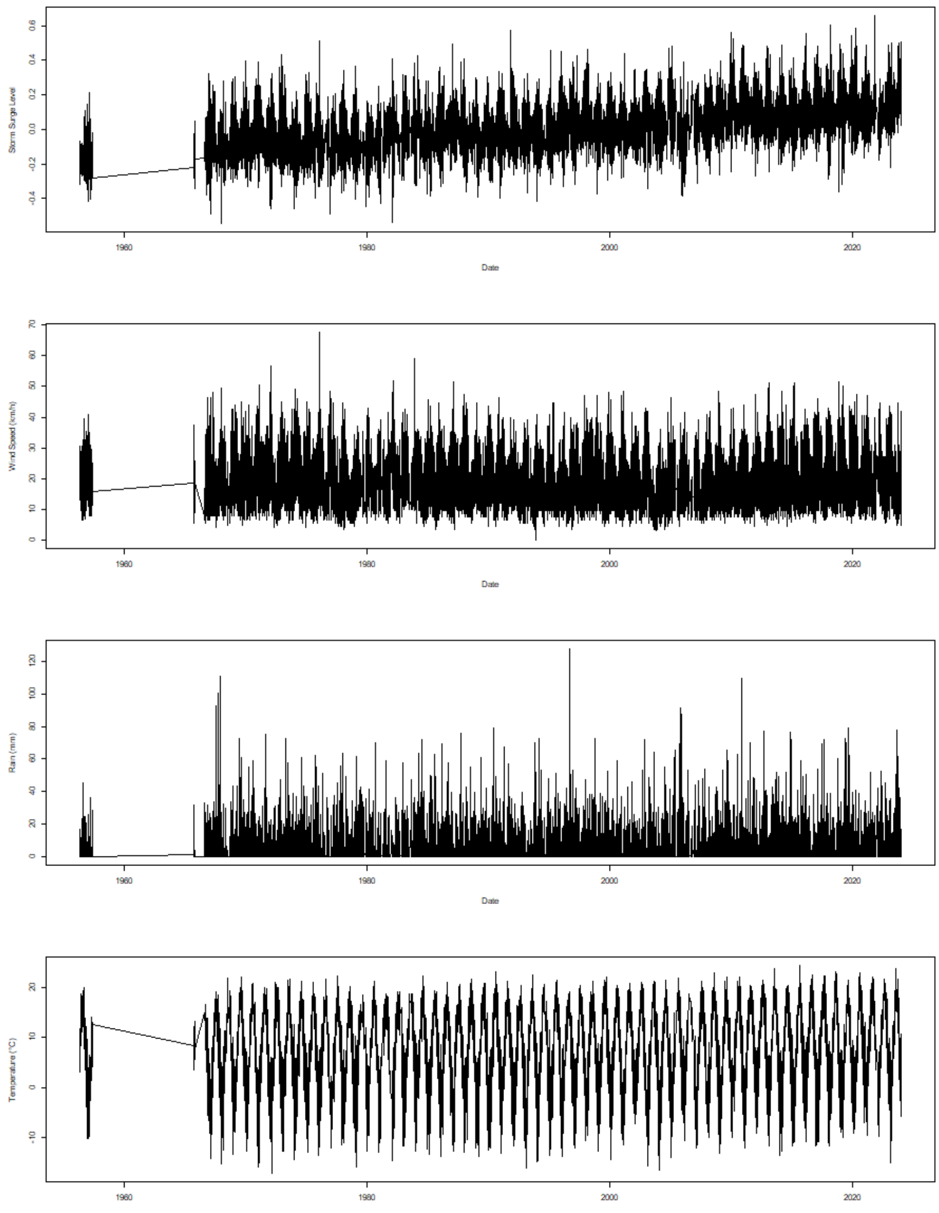

Figure 19: Time series of climatic factors in Yarmouth (storm surge, wind speed, rain, and temperature.)

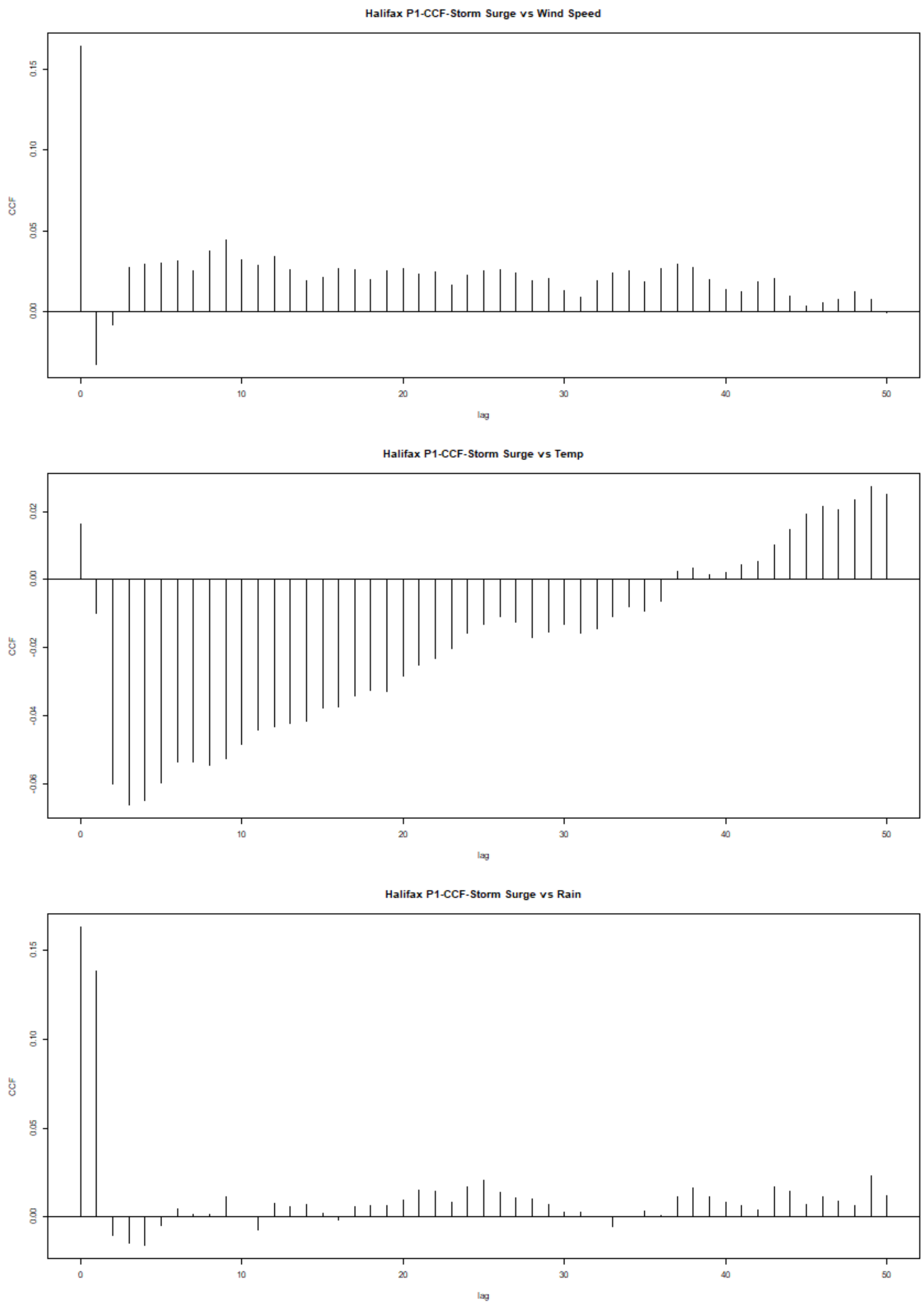

Figure 20: CCF plots of storm surge vs wind speed (top), storm surge vs temperature (middle), storm surge vs rain (bottom) in Yarmouth.

ADF test for each time series for Yarmouth location

```
$avg_water_surge

    Augmented Dickey-Fuller Test

data:  newX[, i]
Dickey-Fuller = -17.551, Lag order = 26, p-value = 0.01
alternative hypothesis: stationary


$avg_temperature

    Augmented Dickey-Fuller Test

data:  newX[, i]
Dickey-Fuller = -8.4732, Lag order = 26, p-value = 0.01
alternative hypothesis: stationary


$avg_wind_speed

    Augmented Dickey-Fuller Test

data:  newX[, i]
Dickey-Fuller = -13.361, Lag order = 26, p-value = 0.01
alternative hypothesis: stationary


$rain

    Augmented Dickey-Fuller Test

data:  newX[, i]
Dickey-Fuller = -24.346, Lag order = 26, p-value = 0.01
alternative hypothesis: stationary
```

VAR(7) model output for Yarmouth

```
                       Estimate   Std. Error    t value      Pr(>|t|)
avg_water_surge.l1  0.607038522 7.762422e-03  78.202209  0.000000e+00
avg_wind_speed.l1  -0.003132637 9.346235e-05 -33.517636 1.552570e-239
avg_wind_speed.l2   0.001797234 1.043837e-04  17.217584  6.008492e-66
rain.l1             0.001322609 8.298232e-05  15.938447  7.819339e-57
```

```
avg_temperature.l2 -0.004856328 3.086295e-04 -15.735139  1.902118e-55
rain.l2            -0.001226668 8.376249e-05 -14.644596  2.629918e-48
avg_temperature.l1  0.003328524 2.401642e-04  13.859366  1.793093e-43
avg_water_surge.l7  0.080255923 7.735126e-03  10.375516  3.724910e-25
avg_temperature.l3  0.001809273 3.126276e-04   5.787312  7.261540e-09
avg_water_surge.l2  0.045438692 9.465355e-03   4.800527  1.594187e-06
```

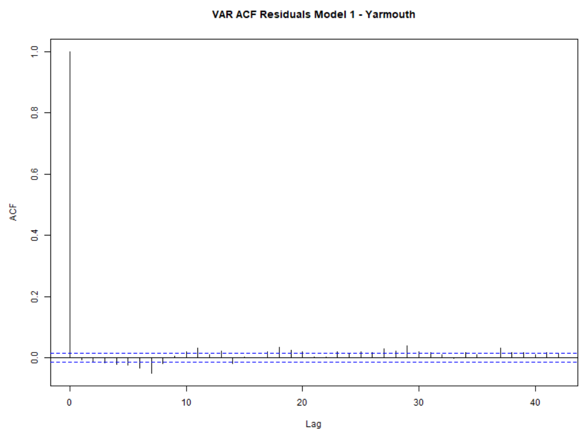

Figure 21: ACF plot of residuals of VAR(7) model for Yarmouth.

VAR(1) model output of water surge and wind speed for Yarmouth

```
Call:
lm(formula = y ~ -1 + ., data = datamat)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.66804 -0.04307  0.00409  0.05518  0.60593 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
avg_water_surge.l1  7.088e-01  5.063e-03   140.0   <2e-16 ***
avg_wind_speed.l1  -3.652e-04  3.381e-05   -10.8   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09148 on 19579 degrees of freedom
Multiple R-squared:  0.5003,    Adjusted R-squared:  0.5002 
F-statistic:  9800 on 2 and 19579 DF,  p-value: < 2.2e-16
```

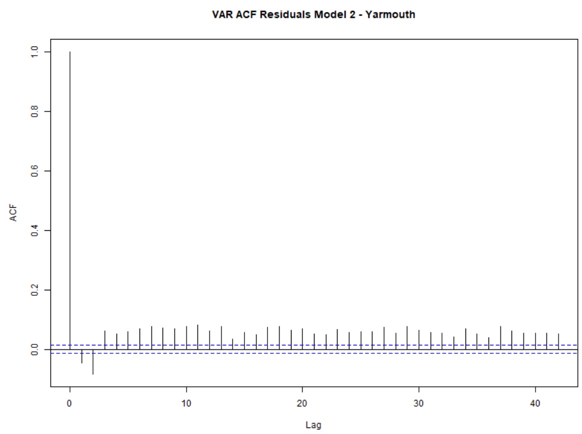

Figure 22: ACF plot of residuals of VAR(1) model of water surge and wind speed in Yarmouth.

AR(1) model output of water surge level in Yarmouth

```
Call:
arima(x = yarmouth_var$avg_water_surge, order = c(1, 0, 0))

Coefficients:
         ar1  intercept
      0.7053     0.0006
s.e.  0.0051     0.0022

sigma^2 estimated as 0.00842:  log likelihood = 18986.65,  aic = -37967.3

Training set error measures:
                       ME       RMSE        MAE      MPE     MAPE      MASE
Training set 1.327234e-05 0.09176235 0.06674873 316.2879 537.3723 0.9440958
                    ACF1
Training set -0.04610327

[1] "R-squared: 0.4973"
```

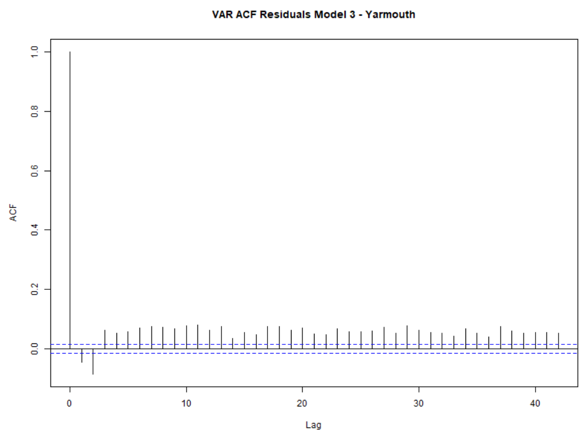

Figure 23: ACF plot of residuals of AR(1) model of storm surge level in Yarmouth.

Granger causality test for Yarmouth location:

```
[1] "All covariates"


    Granger causality H0: avg_temperature avg_wind_speed rain do not
    Granger-cause avg_water_surge

data:  VAR object yarmouth_var_model
F-Test = 70.238, df1 = 3, df2 = 78308, p-value < 2.2e-16


[1] "Wind speed"


    Granger causality H0: avg_wind_speed do not Granger-cause
    avg_water_surge

data:  VAR object yarmouth_var_model_wind
F-Test = 116.71, df1 = 1, df2 = 39158, p-value < 2.2e-16


[1] "Temperature"


    Granger causality H0: avg_temperature do not Granger-cause
    avg_water_surge
```

```
data:  VAR object yarmouth_var_model_temp
F-Test = 8.3674, df1 = 1, df2 = 39158, p-value = 0.003822


[1] "Rain"


    Granger causality H0: rain do not Granger-cause avg_water_surge

data:  VAR object yarmouth_var_model_rain
F-Test = 20.206, df1 = 1, df2 = 39158, p-value = 6.972e-06
```

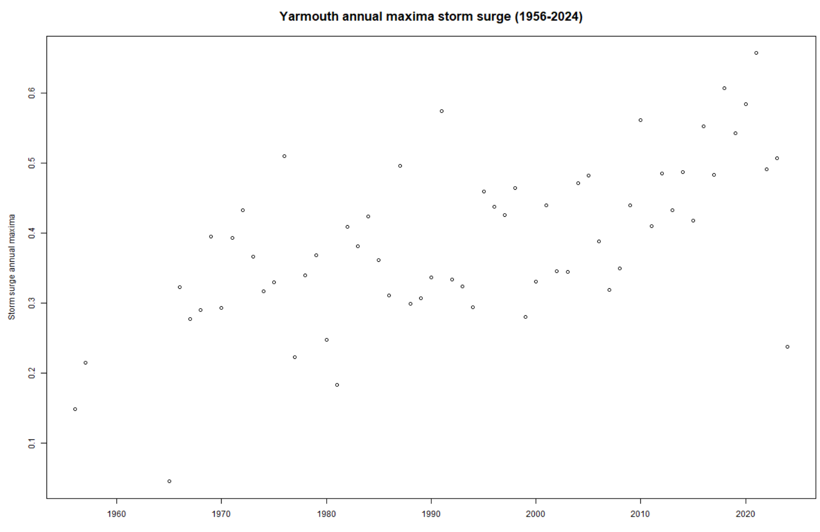

Figure 24: Yarmouth Annual Maximas Plotted against time

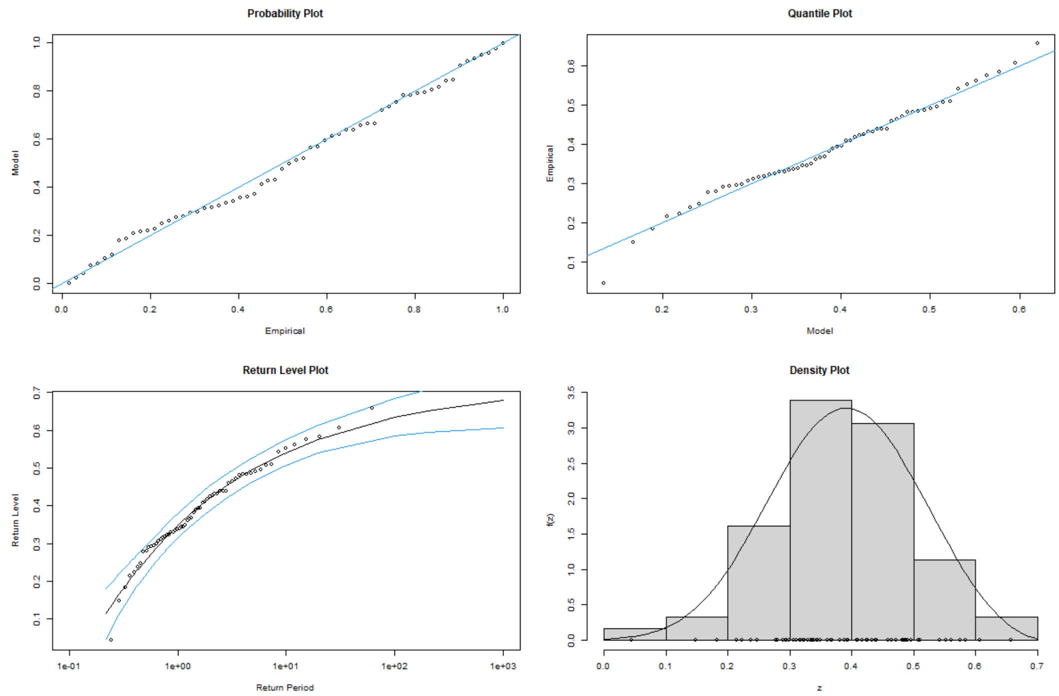

Figure 25: Yarmouth Diagnostics plots from the GEV fit - Non Covariate Case

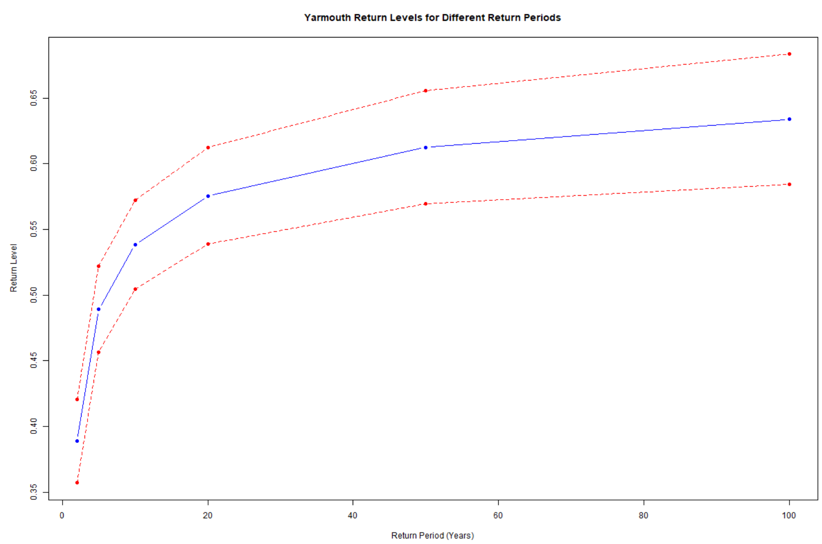

Figure 26: Yarmouth Return Level Plot - Non Covariate Case

Return Period (years)	2	5	10	20	50	100
Storm Surge Level (m)	0.39	0.49	0.54	0.58	0.61	0.63

Table 5: Yarmouth Storm surge levels (in meters) corresponding to various return periods.

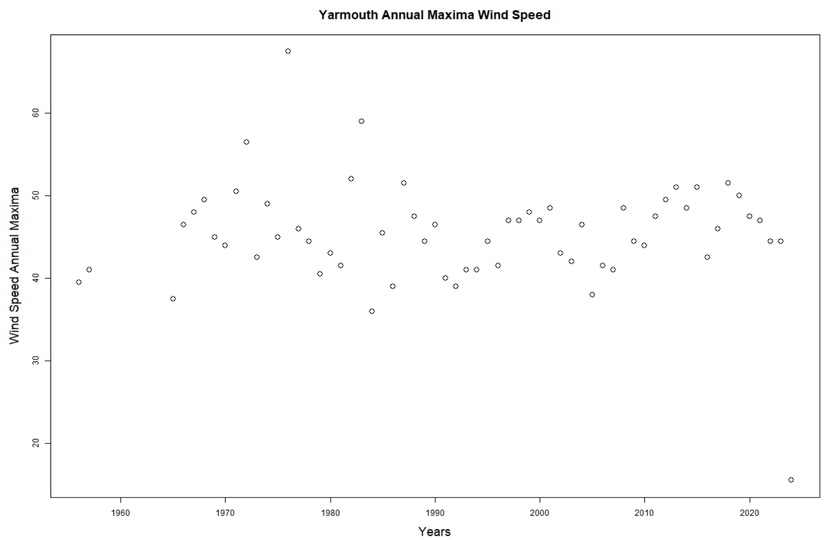

Figure 27: Yarmouth Wind Speed Plot

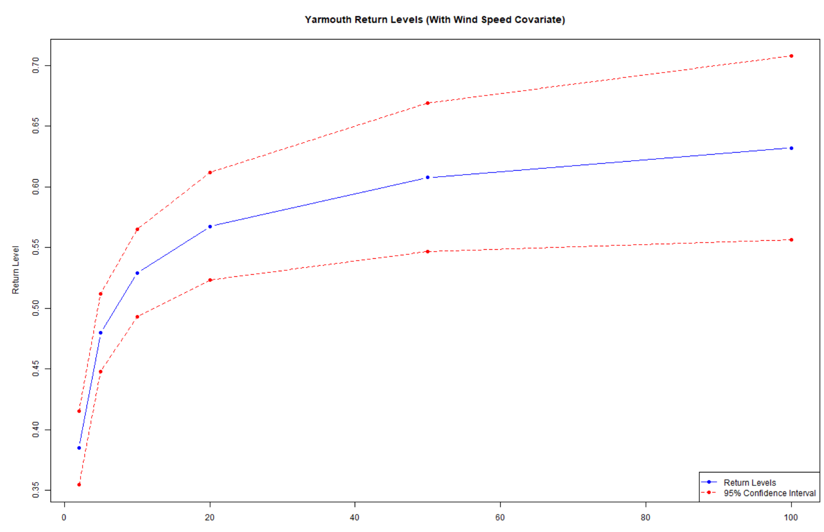

Figure 28: Yarmouth Return Level Plot - Covariate Case

Return Period (years)	2	5	10	20	50	100
Storm Surge Level (m)	0.39	0.48	0.53	0.57	0.61	0.63

Table 6: Yarmouth Storm surge levels (in meters) corresponding to various return periods - Covariate Case.

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